**LATECOMER MANAGEMENT SYSTEM**

***Project Report Submitted to Advanced Technical Training Centre, Sikkim***
***For the partial fulfilment for the award of***

**Diploma in Computer Engineering**
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A.T.T.C

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**BARDANG, EAST SIKKIM-737134**

# DECLARATION

We hereby declare that the project report entitled Late Comers Management System, submitted to the Department of Computer Engineering, ATTC, Bardang, East Sikkim, in partial fulfillment of the requirements for the award of the Diploma in Computer Engineering, is a record of original work carried out by us under the guidance and supervision of Mr. Sonam Palden Barfungpa.

We further declare that this project report has not been submitted earlier, either in part or in full, for the award of any Degree, Diploma, Associateship, Fellowship, or any other similar qualification of any University or Institution.

Dated                                                                Signature of the Candidate

Manzil Pokhrel (A22DCE-17)

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## CERTIFICATE

Certified that the project work entitled Late Comers Management System is a bonafide work submitted by the students mentioned below, in the academic year 2025-2026 for the partial fulfilment for the award of Diploma in Computer Engineering.

Manzil Pokhrel (A22DCE-17)

Signature of Guide                                        Signature of HoD

Signature of External Examiner

# ACKNOWLEDGEMENT

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Manzil Pokhrel - A22DCE-17

# ABSTRACT

The Late Comers Management System (LCMS) is a web-based application designed for schools and colleges. It automates and modernises the process of monitoring late student arrivals each day. Traditional educational institutions often track student punctuality manually without any digital support. Gatekeepers usually maintain handwritten registers to record each student's entry time. That manual process is not only time-consuming but also labour-intensive for the staff. Handwritten records are prone to errors, omissions, and even intentional manipulation by students.

The proposed LCMS directly addresses all these challenges through a digital approach. It provides a digital, barcode-driven solution that eliminates paperwork and guesswork. The system records every student's entry time in real time as they scan. It automatically flags late arrivals based on a configurable threshold time set by administrators. The system is built using standard web technologies for reliability and ease of maintenance. PHP handles all server-side processing, including login, data validation, and business logic. MySQL manages the database, storing student records, attendance logs, and user roles. HTML, CSS, and JavaScript together build the front-end interface that users interact with. The entire application is deployed on a local XAMPP server environment for cost-effective hosting.

Each student is issued a unique barcode-printed ID card for quick identification. These ID cards are scanned at the institution's gate using a standard barcode reader. The LCMS supports four distinct user roles to separate access and responsibilities. The four roles are: Administrator, Faculty, Student, and Gatekeeper. Each role has a dedicated dashboard tailored to its specific tasks and needs. Each role also has specific privileges, such as viewing, editing, or generating reports.

A built-in notification mechanism automatically alerts the respective Section In-charge. Alerts are triggered when a student accumulates three or more late arrivals in a period. Extensive testing was carried out before final deployment to ensure reliability. Testing included unit, integration, system, and user acceptance testing (UAT) phases. The average time from barcode scan to record completion is under two seconds. The system

proved to be accurate, scalable, and suitable for large student populations. It works flawlessly with all major web browsers, including Chrome, Firefox, and Edge.
LCMS improves discipline management effectively by providing real-time late arrival data. It significantly reduces administrative overhead by automating recording and notifications. It creates a transparent attendance ecosystem where students and faculty can track punctuality. Future enhancements planned for the system include SMS notifications for parents. Also planned are a mobile application development and biometric authentication support.

**Keywords:** Latecomers Management, Barcode Attendance, PHP, MySQL, XAMPP, Role-Based Access Control, Web Application, Student Discipline.

# LIST OF ABBREVIATIONS

| Acronym | Full Form |
|---|---|
| LCMS | Late Comers Management System |
| ATTC | Advanced Technical Training Centre |
| AICTE | All India Council for Technical Education |
| PHP | Hypertext Preprocessor |
| MySQL | My Structured Query Language |
| XAMPP | Cross-Platform Apache MySQL PHP Perl |
| HTML | Hypertext Markup Language |
| CSS | Cascading Style Sheets |
| JS | JavaScript |
| CRUD | Create, Read, Update, Delete |
| RBAC | Role-Based Access Control |
| DFD | Data Flow Diagram |
| ER | Entity-Relationship |
| SRS | Software Requirements Specification |
| DPR | Detailed Project Report |
| UAT | User Acceptance Testing |
| PDF | Portable Document Format |
| XLSX | Excel Spreadsheet Format |
| AJAX | Asynchronous JavaScript and XML |
| SHA | Secure Hash Algorithm |
| RFID | Radio Frequency Identification |
| QR | Quick Response |
| API | Application Programming Interface |
| HOD | Head of Department |
| SDLC | Software Development Life Cycle |
| LAN | Local Area Network |
| SMS | Short Message Service |
| SMTP | Simple Mail Transfer Protocol |

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# CHAPTER 1: INTRODUCTION

## 1.1 Background

Educational institutions across the world face ongoing challenges in maintaining student discipline every day. Punctuality is one of the most important aspects of overall student discipline and order. Late arrival of students disrupts ongoing classroom activities and disturbs both teachers and peers. It also affects the learning environment negatively by breaking the flow of lessons. It reflects poorly on institutional discipline as well, damaging the institution's reputation. In many colleges and polytechnics, managing late-coming students remains a recurring and stubborn issue. This recurring issue requires proper monitoring and control through structured systems.

At ATTC, Bardang, East Sikkim, recording late-coming students is done manually without automation. Gatekeepers use physical registers made of paper for this recording purpose. Students arriving after the prescribed reporting time of 08:45 AM provide their details verbally. Those details are then written in the register by the gatekeeper manually. This traditional manual method has been used for many years at this institution. However, this approach has several limitations that affect efficiency and accuracy. The manual process is time-consuming because each entry takes several minutes. It is prone to human errors such as misspelling names or wrong times. It is difficult to maintain over time as registers get lost or damaged. It lacks real-time access for faculty and administration who need immediate data. Retrieving previous records is also difficult without search functionality. Preparing reports becomes inefficient and often requires manual counting of entries.

The idea of LCMS was developed after observing these problems firsthand at ATTC. Discussions were held with gatekeepers, faculty members, and administrative staff together. It was found that a computerized system could improve efficiency dramatically over manual methods. It could also improve accuracy of late-entry management by eliminating handwriting errors. Therefore, LCMS was proposed as a web-based application accessible through browsers. It automates and streamlines the entire workflow of recording and monitoring late arrivals.

The system allows gatekeepers to enter student details digitally using a simple form. Administrators and faculty can access records instantly from any connected computer. Access is through a centralized database that stores all entries securely. The LCMS reduces paperwork significantly by replacing physical registers with digital records. It saves time for everyone involved, from gatekeepers to administrators. It improves accuracy of attendance records by validating data at entry time. It helps identify habitual latecomers through automatically generated periodic reports. The system uses PHP, HTML, CSS, JavaScript, and MySQL as its core technologies. It aims to provide a secure, user-friendly solution that staff can learn quickly. It also aims to be efficient for maintaining discipline and punctuality institution-wide.

## 1.2 Problem Statement

The existing manual system for tracking late-arriving students at ATTC faces several operational challenges daily. These challenges are both operational (day-to-day) and administrative (related to management and reporting). They reduce efficiency and accuracy in maintaining attendance discipline across all student cohorts. The current process relies heavily on handwritten registers made of paper and ink. It also relies on manual monitoring by gatekeepers without any digital tools. This becomes difficult to manage during busy entry hours in the morning rush. Many students arrive simultaneously at that time, especially between 8:30 and 9:00 AM. As a result, the system often fails to provide accurate information about late arrivals. It also fails to provide timely information to faculty who need it immediately. Administrators also suffer from this lack of timely and accurate attendance information.

**a) Time-Consuming Process:**

In the manual system, security personnel record each late student's details individually by hand. They record names, roll numbers, departments, and exact arrival times in a register. This recording is done in a register without any digital assistance or automation. During peak entry periods, this process becomes slow, chaotic, and poorly organized. It leads to long queues at the gate and confusion among students waiting. Since entries are done manually, some students may be missed entirely in the rush. Others may provide incorrect information either intentionally or accidentally. This consumes valuable time that could be used for more productive tasks. That time could be used for academic activities like teaching or studying. It could also be used for administrative activities like planning and record keeping.

**b) Inaccurate Records:**

Handwritten records are highly vulnerable to human error because they lack validation checks. Mistakes include incorrect spelling of student names due to fast writing or poor hearing. Also, wrong roll numbers, incomplete details, and illegible handwriting are common problems. These errors are common especially when many students arrive at the same time. Over time, inaccuracies create unreliable data that cannot be trusted for decisions. This affects attendance monitoring and disciplinary actions that depend on accurate records. Manual systems also increase duplicate entries where the same student is recorded twice. Accidental omissions are also more likely because staff may skip pages or names. Overall record management becomes inefficient and frustrating for everyone involved.

**c) Lack of Real-Time Visibility:**

The existing system does not provide instant access to late-arrival information for any user. Faculty members cannot monitor student punctuality in real time without visiting the gate. Section in-charges also cannot monitor in real time because registers are not shared digitally. Administrators cannot monitor either unless they physically examine the handwritten register. Records remain confined to physical registers that stay at the gatekeeper's desk. This delay in accessing information makes it difficult to respond quickly to issues. It is difficult to identify patterns of repeated late attendance without reviewing many pages. It prevents timely intervention for students who need counselling or warnings. It also prevents counselling for punctuality issues that could be corrected early.

**d) No Automated Alerts or Notifications:**

In the current manual process, there is no automatic notification mechanism built into the system. Section in-charges are not alerted automatically when a student accumulates multiple late arrivals. Mentors are not alerted automatically, so they remain unaware of attendance problems. Faculty members are not alerted automatically even when the same student is late repeatedly. This happens when a student repeatedly arrives late over several days or weeks. Staff must manually review registers to identify such cases of chronic lateness. This manual review is time-consuming because registers contain hundreds of entries. It is often neglected due to workload from other teaching or administrative duties. As a result, repeated late arrivals may continue unnoticed for long periods. This affects discipline and overall attendance management across the institution.

**e) Difficult Report Generation:**

Preparing reports from handwritten registers requires significant manual effort and time. Reports can be weekly, monthly, or semester-wise depending on administrative needs. Staff must search through multiple pages of registers to find relevant entries. They must count entries manually because no automated counting feature exists. They must calculate totals manually, adding numbers with pen and paper. This increases the chances of calculation errors due to fatigue or distraction. Generating reports for meetings becomes lengthy, often taking hours or days. Reports for disciplinary actions also become lengthy and delay necessary actions. Reports for administrative reviews are similarly inconvenient and prone to mistakes.

**f) No Secure Digital Archive:**

Physical attendance registers are vulnerable to damage from water, fire, or tearing. They are also vulnerable to loss, theft, or tampering by unauthorized persons. Registers may deteriorate over time due to improper storage in humid or dusty rooms. Environmental conditions like heat, humidity, or pests also cause deterioration. Frequent handling causes wear and tears such as torn pages or faded ink. Since there is no digital backup, recovering lost information becomes nearly impossible. This creates serious issues in maintaining long-term attendance records for years. It also harms institutional transparency because records cannot be reliably audited.

## 1.3 Objectives

The design and development of LCMS were guided by several important objectives established at the start. These objectives aim to improve attendance management and student punctuality across all departments. The system focuses on automation, accuracy, security, and efficient monitoring of attendance data. It focuses on monitoring of late arrivals specifically, which is a common problem in institutions. By replacing the manual process with a digital system, LCMS helps reduce paperwork significantly. It also improves record management by organizing all entries in a structured database. It simplifies attendance monitoring as well because staff can view records instantly.

a. To design and develop a web-based application that automates the recording of student late entries using barcode scanning. To design and develop a web-based application accessible from any modern web browser. It automates recording of student late entries without any manual writing or register use. It uses barcode scanning for this purpose, which is fast and reliable. The system uses barcode scanning technology to quickly record entries as students enter. This reduces manual work that gatekeepers previously had to do by hand. It saves time during peak entry hours when many students arrive together. It improves attendance accuracy by eliminating handwriting errors and missed entries.

b. To implement role-based access control for Admins, Faculty, Students, and Gatekeepers:
To implement role-based access control that restricts what each user can see and do. Roles: Admins, Faculty, Students, and Gatekeepers, each with different permissions. Each user gets access only to features related to their responsibilities and job duties. This improves system security by preventing unauthorized access to sensitive data. It ensures proper management of information because users see only what they need.

c. To develop a centralized MySQL database for secure data storage:
To develop a centralized MySQL database that stores all system data in one place. It is for secure data storage with protection against unauthorized access or loss. The database securely stores attendance records for every student over time. It stores student details, user information, and late-entry data in structured

tables. This makes record management easier and more reliable compared to paper registers.

d. To create user-friendly dashboards for different users based on their roles. Separate dashboards are provided for Admins, Faculty, Students, and Gatekeepers. They can easily view and manage relevant information without confusion or extra steps.

e. To implement automatic late detection using the predefined late threshold time (8:45AM):
To implement automatic late detection without any manual judgment by gatekeepers. It uses the predefined late threshold time (8:45 AM) set by the administrator. The system automatically checks the student entry time against this threshold. It marks students as late if they arrive after the specified time of 8:45 AM.

f. To develop an automated notification system for repeated late arrivals:
To develop an automated notification system that sends alerts without human intervention. It is for repeated late arrivals, specifically targeting chronic latecomers. When a student’s late count reaches three or more instances within a period. The system automatically alerts the concerned Section In-charge via the dashboard.

g. To provide report generation features in PDF and Excel formats:
To provide report generation features that save time for faculty and administrators. Reports can be in PDF and Excel formats for printing or further analysis. Faculty members and administrators can generate reports easily with a few clicks. They can download them for analysis and official use in meetings or disciplinary actions.

h. To ensure system security, reliability, and efficient performance:
To ensure system security, reliability, and efficient performance under all conditions. LCMS is designed to provide secure access through login credentials and role restrictions. It provides reliable data management with backup capabilities and error handling. It provides smooth performance under normal institutional usage with multiple concurrent users.

# CHAPTER 2: SRS (Software Requirements Specification)

## 2.1 Introduction

Effective monitoring of student punctuality is a fundamental component of institutional discipline and academic management. Late arrivals disrupt classroom activities, reduce instructional time, and negatively impact both the latecomer and their peers. Despite the availability of digital solutions, many educational institutions, particularly in developing regions, continue to rely on outdated, manual methods for tracking late-coming students. The existing process at the institution under study is completely manual, depending entirely on a gatekeeper who records student details in a physical register. This method introduces significant inefficiencies, including slow data entry, human errors, lack of real-time access, and risks of data loss or tampering. To address these challenges, this chapter presents a comprehensive System Requirements Specification (SRS) for the proposed Late Comers Management System (LCMS). The LCMS is a web-based, automated solution designed to replace the manual paper-based process. It leverages barcode scanning technology, a centralized MySQL database, and role-based dashboards to automate attendance recording, provide real-time monitoring, generate instant reports, and improve institutional discipline. This SRS document serves as the blueprint for system development, detailing the overall description, design, implementation features, and functional and non-functional requirements necessary for successful deployment.

## 2.2 Overall Description

The proposed Late Comers Management System (LCMS) is a web-based application intended to run on the institution's local area network. It is designed to be accessed from any modern web browser on computers or mobile devices, eliminating the need for specialized hardware beyond a standard barcode scanner. The system directly addresses the limitations of the existing manual process. Currently, each morning a gatekeeper stands at the entrance and records the names and roll numbers of students arriving after 08:45 AM in a paper register. At the end of each day, administrative staff manually compile these entries to prepare weekly or monthly summaries. These summaries are then communicated to class teachers and section in-charges via verbal messages or paper-based notices. This manual workflow is time-consuming, error-prone, lacks real-time visibility, and places physical registers at risk of loss, damage, or tampering. In contrast, the LCMS automates the entire workflow from data capture to reporting. Students scan their barcode-based ID cards at the entrance; the system instantly identifies the student and records their entry time. If the time exceeds the 08:45 AM threshold, the system automatically marks the entry as “late” without any human intervention. All records are stored in real time in a centralized MySQL database, providing immediate access to authorized users. The system serves four primary user roles: Gatekeeper (barcode scanning and manual entry), Faculty (monitoring and reports), Student (personal attendance history), and Admin (complete system management). By replacing manual processes with automation, the LCMS significantly improves speed, accuracy, data security, and institutional accountability.

## 2.3 Design and Implementation Constraints

The development of the LCMS followed the Waterfall Software Development Life Cycle (SDLC) model, chosen for its structured, sequential approach where each phase is completed before the next begins. This methodology ensured proper planning, clear documentation, and organized project management. The system architecture is client-server based, with web browsers acting as clients communicating over the network with a central server running the XAMPP stack. XAMPP provides the Apache web server for handling HTTP requests, PHP for server-side scripting and application logic, and MySQL as the relational database management system for secure data storage. The implementation phase utilized PHP for backend operations such as user authentication, attendance processing, and report generation. HTML and CSS were used to structure and style web pages, while JavaScript and Bootstrap added interactivity and responsive design, ensuring usability across different screen sizes. The barcode scanning module was implemented to interface with a standard USB barcode reader. When a student’s ID card is scanned, the system queries the database, identifies the student, and triggers the attendance recording logic. The attendance processing module applies the institutional rule (08:45 AM) to the entry time and stores the record as “On Time” or “Late.” The authentication module enforces role-based access control (RBAC) with passwords hashed using SHA-256 for security. The reporting module allows users to generate filtered attendance reports and export them to Excel format. All modules were integrated step by step, with continuous testing to ensure smooth operation and data consistency.

## 2.4 System Features

The LCMS incorporates a comprehensive set of features designed to address the practical needs of each user category and fully automate latecomer management. The first and most important feature is Automated Barcode Attendance. Students are issued ID cards with unique barcodes. When a student scans their card at the gate, the system completes the attendance entry in less than two seconds, eliminating manual handwriting and significantly reducing human error. The second feature is Role-Based Dashboards. The system provides four distinct interfaces: the Admin Dashboard for managing faculty and gatekeeper accounts, assigning section in-charges, and viewing system analytics; the Faculty Dashboard for monitoring attendance of assigned sections, generating reports, and receiving notifications; the Student Dashboard for students to view their personal late entry history, entry times, and overall late percentage; and the Gatekeeper Dashboard for barcode scanning and manual roll-number entry. The third feature is Automated Notifications. The system continuously monitors each student’s late arrival count. When a student accumulates three or more late arrivals, the system automatically sends a notification to the respective Section In-charge, enabling timely disciplinary action. The fourth feature is Centralized Database and Reporting. All attendance records are stored securely in a MySQL database, creating a permanent digital archive safe from physical damage or loss. Authorized users can generate accurate reports filtered by date, student, or class and export them to Excel format. The fifth feature is a Manual Entry Fallback. In cases where a student forgets or loses their ID card, the gatekeeper can manually record a late entry by typing the student’s roll number, ensuring no student is left unrecorded. Together, these features create a complete, efficient, and reliable attendance management solution.

## 2. 5 Functional Requirement

Functional requirements specify the exact behaviours, operations, and tasks that the LCMS must perform. Each requirement is identified with a unique code and described clearly:

| SL. no | Tables |
|---|---|
| 01 | (User Authentication): The system shall require all users (Admin, Faculty, Student, Gatekeeper) to log in using valid credentials. It must block all unauthorized access attempts and differentiate user roles after successful login. |
| 02 | (Student Barcode Management): The system shall assign a unique barcode to each student at the time of registration. The Admin must be able to perform CRUD (Create, Read, Update, Delete) operations on student information linked to their barcodes |
| 03 | (Barcode Scanning): The system shall allow the Gatekeeper to scan a student's barcode using a USB barcode reader. Upon scanning, it must automatically identify the student and record the current date and time as the entry time. |
| 04 | (Late Attendance Recording): The system shall automatically compare the recorded entry time against the institutional threshold of 08:45 AM. If the entry time exceeds this threshold, the record shall be marked as "Late." Each record must store student ID, date, time, and late status. |
| 05 | 05 (Manual Entry Fallback): The system shall provide the Gatekeeper with a form to manually record a late entry by entering the student's roll number when a barcode is unavailable. |
| 06 | (Database Management): All records shall be stored in a centralized MySQL database. The system shall support CRUD operations for authorized users (Admins and Faculty) to maintain data integrity. |
| 07 | (Report Generation): The system shall allow Faculty and Admin users to view and download attendance reports. Reports must be filterable by date range, specific student, or class section, and exportable to Excel format. |
| 08 | (Automated Notifications): The system shall automatically monitor each student's late count per month. When a student accumulates three or more late arrivals, the system shall send an email or in-system notification to the assigned Section In-charge. |
| 09 | (Admin Dashboard): The system shall provide a dashboard for the Admin to manage Faculty and Gatekeeper accounts, assign Section In-charges to classes, and view system-wide attendance analytics. |
| 10 | (Student Dashboard): The system shall provide a dashboard where individual students can view their personal late entry history, daily entry times, and cumulative late percentage. |

## 2. 6 Non -Functional Requirement

Non-functional requirements define the quality attributes, system constraints, and performance benchmarks that the LCMS must meet to ensure a satisfactory user experience and reliable operation.

| SL.no | Tables |
|---|---|
| 01 | **Performance:** The system shall complete each attendance entry within two seconds of a barcode scan, from the moment the barcode is read to the confirmation of database storage. Report generation queries involving standard filters (date range, single class) shall execute within five seconds. |
| 02 | **Usability:** The user interface shall be intuitive, with clear navigation and consistent layout across all dashboards. The system shall require minimal training for gatekeepers and faculty members, with on-screen instructions and tooltips provided where needed. |
| 03 | **Reliability**: The system shall ensure accurate recording of attendance without data duplication, loss, or corruption. It must handle concurrent scans during peak morning hours (e.g., multiple students arriving at the same time) without crashing or mixing records. |
| 04 | **Security:** The system shall enforce role-based access control (RBAC), ensuring that users can only access features and data permitted by their role. All user passwords shall be hashed using the SHA-256 algorithm before storage in the database. Session management shall timeout after 15 minutes of inactivity. |
| 05 | **Scalability:** The database schema and application code shall be designed to handle an increasing number of students (from hundreds to thousands) and attendance records (thousands per day) without significant degradation in response time or performance. |
| 06 | **Maintainability**: The source code shall be modular, well-commented, and follow consistent naming conventions. Database queries shall be written to allow easy modification of attendance rules (e.g., changing the 08:45 AM threshold) without rewriting core logic. |

| | |
|---|---|
| 07 | **Availability:** The system shall guarantee 99% uptime during official college working hours (typically 8:00 AM to 4:00 PM). Planned maintenance shall be scheduled outside these hours. |
| 08 | **Compatibility:** The system shall be fully functional and display correctly on all major modern web browsers, including the latest stable versions of Google Chrome, Mozilla Firefox, and Microsoft Edge. It shall also be responsive for access on tablet and mobile devices. |

## 2. 7 Methodology

The development of the Late Comers Management System (LCMS) was carried out using the Waterfall Software Development Life Cycle (SDLC) model. This model was selected because it provides a structured, systematic, and sequential approach where each phase must be fully completed before the next phase begins. The Waterfall methodology is highly suitable for academic and institutional projects with clearly defined, stable requirements, as it ensures proper planning, clear documentation, organized development, and easier project management. The six phases were applied as follows.

| SL.no | Tables |
|---|---|
| 01 | **Phase 1** - Requirements Gathering and Analysis: In this initial phase, the limitations of the existing manual system were studied through direct observation and discussions with gatekeepers, faculty, and administrative staff. Both functional requirements (barcode scanning, attendance recording, report generation) and non-functional requirements (security, performance, usability) were identified and documented. |
| 02 | **Phase 2** - System Design: Based on the requirements, the overall system architecture was planned. Data Flow Diagrams (DFDs) were created to map how attendance data moves between users, processes, and the database. Entity Relationship (ER) diagrams were designed to define database tables |

| | |
|---|---|
| | (students, users, attendance records) and their relationships. User interface layouts for each role-based dashboard were also designed**.** |
| 03 | **Phase 3** - Implementation (Coding): Using PHP, MySQL, HTML, CSS, JavaScript, and Bootstrap on the XAMPP platform, all modules were coded and integrated. The barcode scanning, attendance processing, authentication, dashboards, notifications, and reporting modules were developed according to the design specifications. |
| 04 | **Phase 4 -** Testing: The system was thoroughly tested to identify and correct errors. Unit testing verified individual components (e.g., login, barcode scan). Integration testing ensured smooth communication between modules. System testing evaluated the complete application's functionality and performance. User Acceptance Testing (UAT) involved actual users (gatekeepers, faculty) to confirm the system met their practical needs. All identified bugs were fixed. |
| 05 | **Phase 5** - Deployment: After successful testing, the LCMS was deployed on the institution's local XAMPP server environment. The barcode scanner was configured at the gatekeeper's workstation. User accounts and role-based permissions were created. The system was made fully operational for daily attendance management. |
| 06 | **Phase 6** - Maintenance: Ongoing maintenance includes regular monitoring for bugs, performance optimization, weekly database backups, and minor feature enhancements based on user feedback. User training and system documentation were also provided to ensure long-term stable operation. This disciplined Waterfall approach contributed significantly to the successful completion of the project. |

# CHAPTER 3: DESIGN & DEVELOPMENT

## 3.1 Design

The LCMS follows a three-tier client-server architecture that separates responsibilities clearly. It separates the system into different layers for modularity and easier maintenance. This is for better performance, security, and management of the entire system. This architecture allows smooth communication between all components without delays. Communication happens between the user interface, application logic, and database seamlessly.

i. Presentation Tire (Tier):

   The Presentation Tier is the front-end layer that users see and interact with. Users interact with it through a web browser on any device. It is developed using HTML, CSS, and JavaScript for responsiveness and visual design. It provides login pages and dashboards tailored to each user role. It provides barcode scanning interfaces for fast student identification at the gate. It provides attendance records and report pages for viewing and downloading data. These are for different users such as Admins, Faculty, Students, and Gatekeepers.

ii. Application Tire (Tier):

   The Application Tier contains the business logic of the system, meaning all rules and calculations. It is developed using PHP, a server-side scripting language. It handles important functions such as user authentication by verifying login credentials. It handles barcode processing to decode student ID numbers from scans. It handles attendance recording by saving entry times to the database. It handles automatic late detection by comparing entry times with the 08:45 AM threshold. It handles notification generation when a student accumulates three or more late arrivals. It handles report management by compiling attendance data into Excel files.

iii. Data Tire (Tier):

The Data Tier consists of the MySQL database where all persistent data lives. It securely stores all system data with protection against unauthorized access. Data includes student details, attendance records, and user information like passwords and roles. It also includes notifications and reports generated by the application tier. It ensures centralized and reliable data management with backup and recovery capabilities.

### 3.1.2 Data Flow Diagrams (DFD)

Context-Level DFD (Level 0):

The Context-Level DFD provides a high-level overview of the LCMS without internal details. It shows the system as a single integrated process that handles all attendance functions. The system interacts with four main external entities that are outside the system boundary. These entities are: Admin, Faculty, Student, and Gatekeeper, each with different roles. These entities exchange data with the system in both directions as needed. Data includes login credentials, barcode scan data, and attendance records for late arrivals. Also, notifications and report requests that flow between users and the system. The diagram shows how information flows between users and the LCMS using data flow arrows. It presents the system as one central process without showing internal sub-processes.

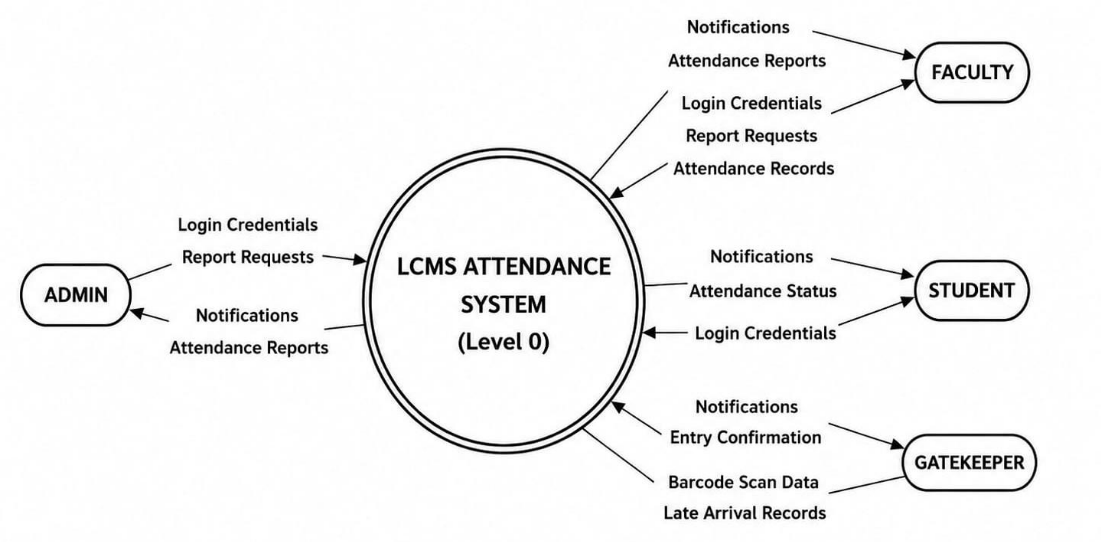



**Fig 3.1**

**Level 1 DFD - Admin Module:**

The Level 1 DFD for the Admin Module describes the major processes performed by the administrator in detail. These include Authenticate Admin, Manage Users, Assign In-charge, and View Analytics for attendance data. The admin can create and manage user accounts for faculty, gatekeepers, and students. The admin can assign section in-charges to specific academic sections or departments. The admin can monitor attendance statistics, including late arrival trends and repeat offenders. The module interacts with data stores such as Users, Sections, and Attendance Log for persistence. It stores and retrieves system information from these data stores as needed for operations.

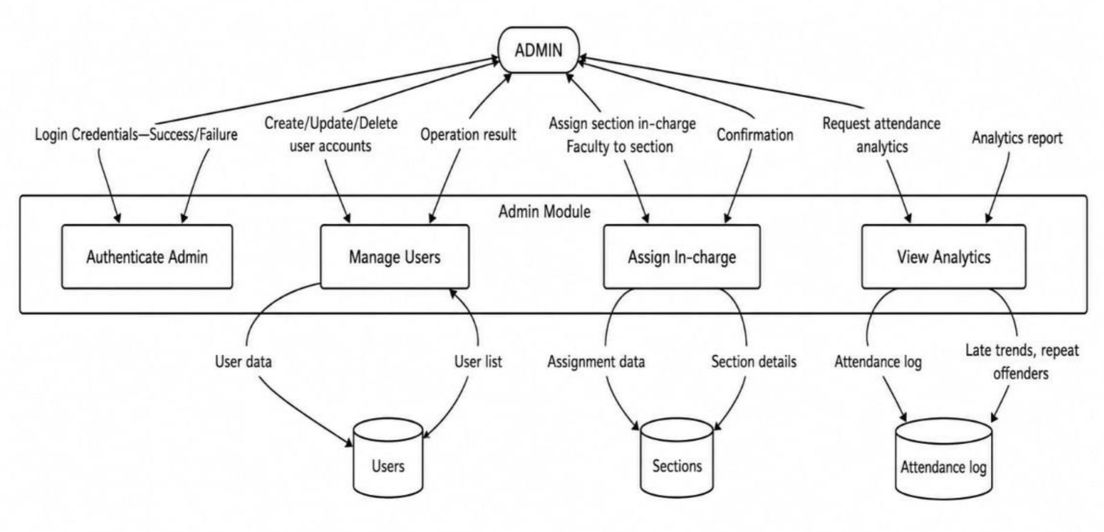



**Fig 3.2**

**Level 1 DFD - Gatekeeper Module:**

The Gatekeeper Module handles the process of recording late student entries at the institution’s entrance. Its main processes include Authenticate Gatekeeper, Scan Barcode, Record Attendance, and Manual Entry as backup. When a student barcode is scanned, the system verifies the student information against the database. It records the attendance automatically after verification, marking late if after 08:45 AM. This module mainly interacts with the Students and Attendance Log data stores for data retrieval and storage.

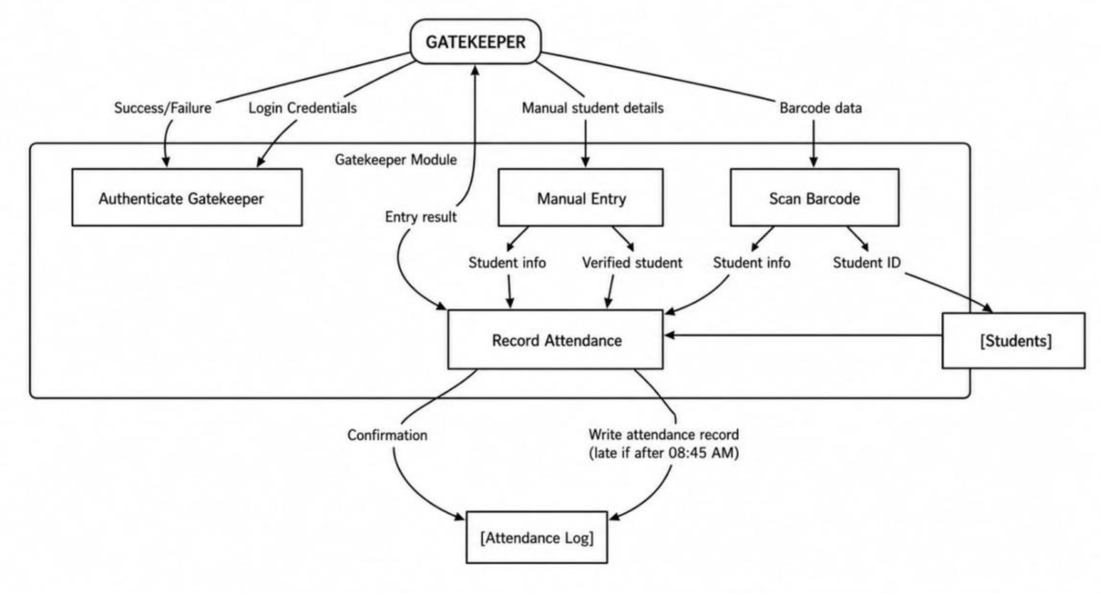



**Fig 3.3**

**Level 1 DFD - Faculty Module:**

The Faculty Module allows faculty members or section in-charges to monitor student attendance daily. It also allows them to receive notifications about repeated late arrivals from the system automatically. The processes include Authenticate Faculty, Manage Students, View Reports, Receive Notifications, and Export Reports. The module accesses data from Students, Attendance Log, and Notifications data stores for retrieval. It uses this data to generate reports and monitor student punctuality effectively over time.

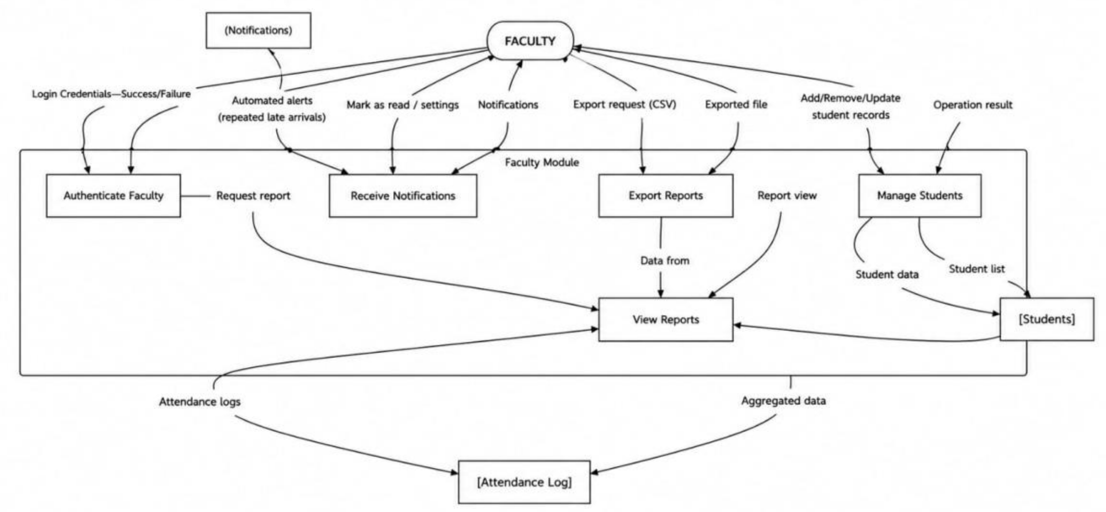



**Fig 3.4**

### 3.1.3 Entity-Relationship (ER) Diagram

The ER diagram includes the following entities:

a) User (user_id PK, uid, pwd, role, log_date,log_time)

b) Student (sid PK, rollno, fname,mname,lname,did,fathername,mothername, mobile,email_id,pwd,gender, semester, sem_year, log_date,log_date)

c) Teacher (tid PK, tfname, tmname, tlname, did,tmobile,hire_date,taddress,status,hod, email_id FK)

d) Gatekeeper (gid PK, gfname,,glname,email_id,hire_date,status,gaddress,gmobile, email_id FK)

e) Attendance (att_id PK, rollno FK, status, date)

f) Administration (aid PK, email_id FK, afname, alname)

g) Dept (did PK, dname,status)

h) Entry_log (entry_log_id PK, rollno FK, log_time,is_late, log_date)

i) No_id (no_id PK, rollno PK, log_date,log_time)

j) Section_incharge (si_id PK, did FK,tid FK, year)

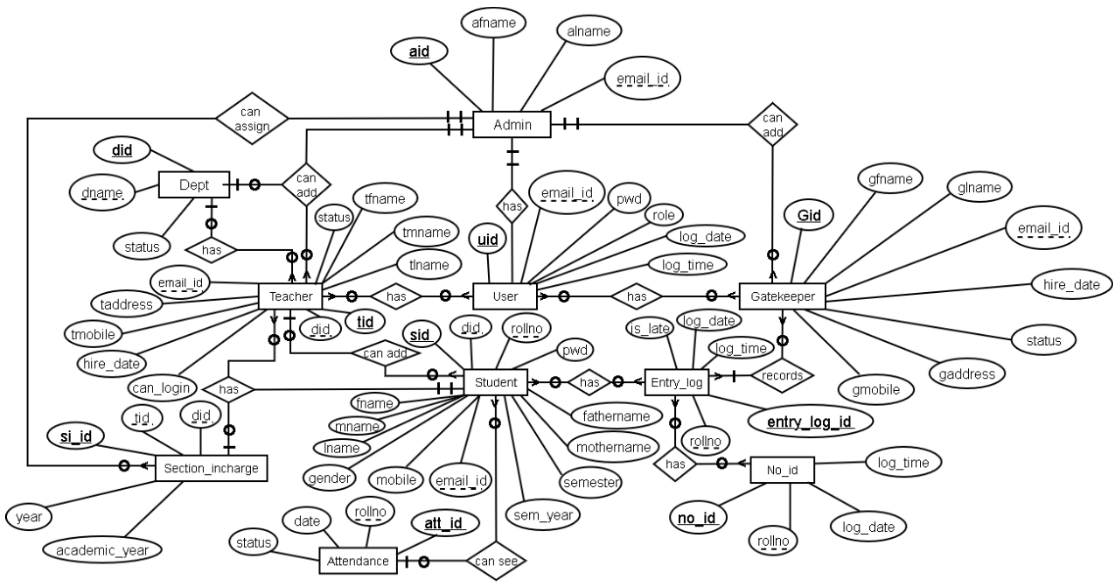



**Fig 3.5**

### 3.1.4 Database Schema

**Table: administrator**

| Column | Type | Description |
|---|---|---|
| aid | INT(11) PK | Primary key, unique administrator ID |
| afname | VARCHAR(255) | Administrator first name |
| alname | VARCHAR(255) | Administrator last name |
| email_id | VARCHAR(255) | Administrator email address |

**Table: attendance**

| Column | Type | Description |
|---|---|---|
| att_id | INT(11) PK | Primary key, attendance record ID |
| rollno | VARCHAR(45) | Student roll number (FK to student) |
| status | VARCHAR(10) | Attendance status value |
| date | DATE | Date of the attendance record |

**Table: dept**

| Column | Type | Description |
|---|---|---|
| did | INT(11) PK | Primary key, department ID |
| dname | VARCHAR(255) | Department name |
| status | ENUM('0','1') | Active/inactive status (default: 1 = active) |

**Table: entry_log**

| Column | Type | Description |
|---|---|---|
| entry_log_id | INT(11) PK | Primary key, entry log record ID |
| rollno | VARCHAR(45) | Student roll number (FK to student) |
| log_time | TIME | Time of student entry at gate |
| is_late | TINYINT(1) | Late flag: 1 = late, NULL = on time |
| log_date | DATE | Date of the entry log |

**Table: gatekeeper**

| Column | Type | Description |
|---|---|---|
| gid | INT(11) PK | Primary key, gatekeeper ID |
| gfname | VARCHAR(255) | Gatekeeper first name |
| glname | VARCHAR(255) | Gatekeeper last name |
| email_id | VARCHAR(255) | Gatekeeper email address |
| hire_date | DATE | Date the gatekeeper was hired |
| status | ENUM('0','1') | Employment status (default: 1 = active) |
| gaddress | VARCHAR(255) | Gatekeeper residential address |
| gmobile | VARCHAR(255) | Gatekeeper mobile contact number |

**Table: no_id**

| Column | Type | Description |
|---|---|---|
| no_id | INT(11) PK | Primary key, record ID |
| rollno | VARCHAR(20) | Student roll number |
| log_date | DATE | Date of the log entry |
| log_time | TIME | Time of the log entry |

**Table: section_incharge**

| Column | Type | Description |
|---|---|---|
| si_id | INT(11) PK | Primary key, section incharge record ID |
| year | YEAR(4) | Academic year (default: current year) |
| tid | INT(11) FK | Teacher ID (FK to teacher) |
| did | INT(11) FK | Department ID (FK to dept) |
| academic_year | ENUM('1st','2nd','3rd') | Academic year level of the section |

**Table: student**

| Column | Type | Description |
|---|---|---|
| Sid | INT(11) PK | Primary key, unique student ID |
| Fname | VARCHAR(255) | Student first name |
| Mname | VARCHAR(255) | Student middle name |
| Lname | VARCHAR(255) | Student last name |
| Rollno | VARCHAR(255) | Unique student roll number (NOT NULL) |

| | | |
|---|---|---|
| Did | INT(11) FK | Department ID (FK to dept) |
| fathername | VARCHAR(255) | Student father's name |
| mothername | VARCHAR(255) | Student mother's name |
| Mobile | VARCHAR(255) | Student mobile contact number |
| email_id | VARCHAR(255) | Student email (login, NOT NULL) |
| Pwd | VARCHAR(255) | Hashed login password (NOT NULL) |
| Gender | VARCHAR(255) | Student gender |
| Semester | VARCHAR(50) | Current semester of the student |
| sem_year | VARCHAR(4) | Year of the current semester |
| log_date | VARCHAR(25) | Record creation date (default: curdate()) |
| log_time | VARCHAR(25) | Record creation time (default: curtime()) |

**Table: teacher**

| Column | Type | Description |
|---|---|---|
| tid | INT(11) PK | Primary key, unique teacher ID |
| tfname | VARCHAR(255) | Teacher first name |
| tlname | VARCHAR(255) | Teacher last name |
| email_id | VARCHAR(255) | Teacher email address |
| did | INT(11) FK | Department ID (FK to dept) |
| tmobile | VARCHAR(15) | Teacher mobile contact number |
| hire_date | DATE | Date the teacher was hired |
| taddress | VARCHAR(255) | Teacher residential address |
| status | ENUM('0','1') | Employment status (default: 1 = active) |
| hod | ENUM('Yes','No') | Head of Department flag (default: No) |
| tmname | VARCHAR(255) | Teacher middle name |
| Column | Type | Description |
| tid | INT(11) PK | Primary key, unique teacher ID |
| tfname | VARCHAR(255) | Teacher first name |
| tlname | VARCHAR(255) | Teacher last name |
| email_id | VARCHAR(255) | Teacher email address |
| did | INT(11) FK | Department ID (FK to dept) |

| | | |
|---|---|---|
| tmobile | VARCHAR(15) | Teacher mobile contact number |
| hire_date | DATE | Date the teacher was hired |

**Table: user**

| Column | Type | Description |
|---|---|---|
| uid | INT(11) PK | Primary key, unique user ID |
| email_id | VARCHAR(255) | User email address (login credential) |
| pwd | VARCHAR(255) | Hashed user password |
| role | VARCHAR(255) | User role (admin/faculty/student/gatekeeper) |
| log_date | DATE | Date of last login |
| log_time | VARCHAR(50) | Time of last login |

### 3.1.5 Interface Design Specifications

**Login Page:**

The Login Page acts as the primary authentication and security gateway of the Late Comers Management System (LCMS). It is the first interface displayed to users when they access the application and is designed to provide a secure, simple, and user-friendly login experience. The page allows authorized users to enter their login credentials, including their username or email ID and password, in order to gain access to the system.

The interface contains essential components such as input fields for username and password, a “Login” button to initiate authentication, and a “Forgot Password” option that enables users to recover or reset their accounts in case they forget their login credentials. Proper validation mechanisms are implemented to ensure that incorrect or incomplete details are not accepted. If invalid credentials are entered, the system displays an appropriate error message, helping users identify and correct mistakes easily.

The Login Page also plays a major role in maintaining the security of the system. Only registered and authorized users are allowed to access the application, preventing unauthorized individuals from viewing or modifying sensitive attendance records.

Password protection and authentication checks help maintain data confidentiality and integrity within the system.

Once the entered credentials are successfully verified, the system automatically redirects users to their respective dashboards based on their assigned roles. For example, administrators are redirected to the Admin Dashboard, faculty members to the Faculty Dashboard, students to the Student Dashboard, and gatekeepers to the Gatekeeper Dashboard. This role-based access control mechanism ensures that each user can only access the features, modules, and information relevant to their responsibilities.

By implementing a secure and organized login mechanism, the Login Page improves system efficiency, enhances data security, and provides a smooth user experience for all categories of users within the Late Comers Management System.

**Gatekeeper Dashboard:**

The Gatekeeper Dashboard is a dedicated module of the Late Comers Management System (LCMS) designed specifically for security personnel or gatekeepers responsible for monitoring student entry at the institution gate. The dashboard is developed with a simple, fast, and user-friendly interface to ensure smooth attendance recording during busy entry hours. Its primary objective is to reduce manual work and enable quick identification of students entering the campus.

One of the main features of the dashboard is the large barcode scanning input field, which directly accepts data from a barcode scanner. This allows gatekeepers to record student attendance instantly without the need for manual typing or paper-based registers. When a student's ID card barcode is scanned, the system automatically retrieves the student's information from the database and processes the attendance record within seconds. This significantly improves speed, accuracy, and operational efficiency during peak entry periods.

After each successful scan, the system immediately displays a confirmation panel containing important student details such as the student's name, roll number, department, entry time, and attendance status. The attendance status is represented using a color-coded indication system for easier understanding and faster decision-making. A green indicator signifies that the student has arrived on time, while a red indicator shows that the student is late according to the predefined institutional timing rules. This visual representation helps gatekeepers quickly identify latecomers without checking records manually.

The dashboard also contains a daily entry log section that automatically records and displays all student entries scanned throughout the day. The log includes details such as student identity, scan time, and attendance condition. This feature helps gatekeepers maintain organized records, monitor student movement efficiently, and review attendance activities whenever required. It also assists administrators and faculty members in verifying attendance information later for reporting or disciplinary purposes.

In addition, the Gatekeeper Dashboard minimizes human error, eliminates the need for handwritten registers, and improves the overall reliability of attendance tracking. By automating the entry management process, the dashboard ensures faster student verification, better monitoring efficiency, and a more secure and organized campus entry system.

**Admin Dashboard:**

The Admin Dashboard serves as the central control and management panel of the Late Comers Management System (LCMS). It is designed specifically for administrators to monitor, control, and manage all major activities within the system through a single integrated interface. The dashboard provides a secure, organized, and user-friendly environment that allows administrators to perform administrative tasks efficiently without requiring technical expertise.

The interface includes a sidebar navigation menu that provides easy access to different system modules such as User Management, Student Management, Section In-charge Assignment, and Manage Entries. Through the User Management module, administrators can create, update, delete, and manage accounts for different users such as faculty members, students, and gatekeepers. The Student Management module allows the admin to maintain detailed student records, including names, roll numbers, departments, semesters, and barcode information.

The Section In-charge Assignment module enables administrators to assign faculty members or staff to specific departments or sections for better supervision and attendance monitoring. The Manage Entries module provides access to all attendance records and late entry data, allowing administrators to verify, edit, filter, or remove records whenever necessary. These modules collectively ensure smooth system administration and centralized control over institutional attendance management activities.

The main area of the dashboard displays summary statistic cards that provide quick and important information at a glance. These cards typically show the total number of registered students, active users, attendance records, daily entries, and total latecomers. This summarized information helps administrators quickly understand the overall system status without manually checking multiple records or reports.

In addition to statistical summaries, the Admin Dashboard also contains graphical data visualization features such as bar charts and analytical graphs. These visual representations display attendance trends, daily or weekly latecomer statistics, and student punctuality patterns in an easy-to-understand format. Such insights help administrators identify frequent latecomers, monitor institutional discipline, and make informed decisions to improve attendance management and student accountability.

The dashboard further improves efficiency by reducing manual paperwork, simplifying record management, and enabling faster access to information. Since all major operations are centralized within a single platform, administrators can save time, reduce human error, and ensure better coordination among different departments. Overall, the Admin Dashboard plays a vital role in maintaining the effectiveness, security, and reliability of the Late Comers Management System.

**Faculty Dashboard:**

The Faculty Dashboard is a dedicated module of the Late Comers Management System (LCMS) designed to assist faculty members and section in-charges in monitoring, managing, and analysing student attendance efficiently. It provides a clean, organized, and user-friendly interface that allows faculty members to access important attendance-related information quickly and perform their responsibilities with greater accuracy and convenience.

One of the key features of the dashboard is its ability to display notifications and alerts for students with repeated late arrivals or irregular attendance patterns. These notifications help faculty members identify students who frequently arrive late and may require counselling, warnings, or additional supervision. By automatically highlighting such cases, the system reduces the need for manual checking of attendance records and helps faculty members take timely action when necessary.

The dashboard also provides navigation options for accessing different attendance management functions. Faculty members can view class-wise attendance records to monitor the punctuality and attendance status of students within their assigned classes or sections. This feature enables them to track daily attendance activities efficiently and identify trends or irregularities in student attendance behaviors.

Another important feature of the Faculty Dashboard is the student search functionality, which allows faculty members to search and view detailed information about individual students quickly. Faculty users can access details such as student name, roll number, department, attendance history, late entry records, and overall attendance performance. This helps in maintaining transparency and simplifies the process of reviewing student attendance information whenever required.

The dashboard also allows authorized faculty members to add, update, or manage student information within the system. This includes editing student details, updating attendance-related records, and ensuring that the stored information remains accurate and up to date. Such features help reduce paperwork and improve the efficiency of attendance management processes.

In addition, the Faculty Dashboard enhances communication and academic supervision by providing quick access to attendance statistics and latecomer reports. Faculty members can easily monitor student punctuality, maintain proper attendance records, and support institutional discipline more effectively. By automating and simplifying attendance-related tasks, the dashboard saves time, minimizes manual errors, and contributes to a more organized academic management system.

**Student Dashboard:**

The Student Dashboard is a personalized module of the Late Comers Management System (LCMS) designed specifically for students to monitor their attendance records, punctuality status, and personal information conveniently. It provides a simple, interactive, and user-friendly interface that allows students to access important attendance-related details quickly without depending on faculty members or administrative staff.

One of the primary features of the dashboard is the display of detailed attendance information. Students can view the total number of late arrivals recorded in the system along with their overall late attendance percentage. This helps students understand their punctuality performance and remain aware of their attendance status throughout the academic session. By regularly checking these statistics, students can identify attendance issues early and take corrective actions to improve their punctuality and discipline.

The dashboard also contains a detailed date-wise log of late entries, which records every instance when a student arrived late to the institution. Each entry typically includes information such as the date, entry time, and attendance status. This feature ensures transparency by allowing students to verify their attendance records and review their attendance history whenever required. It also reduces confusion or disputes regarding attendance data because students have direct access to their own records.

In addition to attendance monitoring, the Student Dashboard allows students to update and manage their personal profile information. Students can modify details such as contact information, password settings, and other permitted profile data, making the system more interactive and user-friendly. This feature ensures that student records remain accurate and updated within the system database.

The dashboard may also provide notifications or alerts related to attendance warnings, repeated late arrivals, or important announcements from faculty or administrators. Such notifications help students remain informed about their attendance condition and institutional policies regarding punctuality and discipline.

By providing direct access to attendance statistics, latecomer records, and personal information, the Student Dashboard promotes transparency, accountability, and self-monitoring among students. It encourages students to maintain better attendance discipline, improves awareness regarding punctuality, and contributes to a more organized and efficient attendance management system within the institution.

## 3.2 Calculations

A. Performance Calculations:

**a) Average Scan-to-Record Time:**
The total time taken for 250 barcode scans was 350 seconds.

$$\text{Average Scan-to-Record Time} = \frac{350}{250} = 1.4 \text{ seconds}$$

Therefore, the average time required to scan and record one attendance entry is **1.4 seconds**.

**b) Report Generation Time (100 Records):**
The system required approximately **50 seconds** to generate a report containing 100 attendance records, including database retrieval and PDF generation.

**c) System Uptime:**
The system uptime percentage was calculated using the formula:

$$\text{System Uptime} = \left(\frac{\text{Total Uptime}}{\text{Total Time}}\right) \times 100$$
$$= \left(\frac{715}{720}\right) \times 100 = 99.3\%$$

Therefore, the system achieved an uptime of **99.3%**, indicating high reliability and continuous availability.

B. Cost Savings Calculation:

**a) Manual Process Time:**
In the manual system, report compilation required approximately 45 minutes per week.

$$45 \text{ minutes} \times 52 \text{ weeks} = 2340 \text{ minutes/year}$$
$$2340 \div 60 = 39 \text{ hours/year}$$

Therefore, the manual process consumed **39 hours of administrative work annually**.

**b) LCMS Report Generation Time:**
Using the Late Comers Management System (LCMS), report generation takes approximately **10 seconds per report**.

$$10 \text{ seconds} = 0.0028 \text{ hours}$$

This significantly reduces the time required for report preparation.

**c) Annual Time Saved:**

$$39 \text{ hours} - 0.0028 \text{ hours} \approx 39 \text{ hours}$$

Therefore, the LCMS saves approximately **39 hours of administrative time annually**, improving efficiency and reducing manual workload.

# CHAPTER 4: ESTIMATION & COSTING

## 4.1 Materials

| Item | Quantity | Purpose |
|---|---|---|
| USB Barcode Scanner | 1 | Scanning student ID cards at gate |
| Desktop/Laptop (existing) | 1 | Hosting XAMPP server and gatekeeper workstation |
| Network Switch/Router (existing) | 1 | Local network connectivity |
| ID Cards with Barcodes | As needed | Student identification |
| Printer (Existing) | 1 | Printing barcode stickers or ID cards |

## 4.2 Cost Analysis

Project Type: Organic Model
Project Size: 15 KLOC
Team Members: 4 Members
Salary: ₹10,000 per month per person

Effort Estimation:

The estimated effort required for the development of the Late Comers Management System (LCMS) was calculated using the Organic Model of software cost estimation. Based on the project size of 15 KLOC, the total development effort was estimated in person-months considering analysis, design, coding, testing, and deployment activities.

Development Time:

The total estimated development time for the project was approximately 11 months. During this period, all project phases including requirement analysis, system design, implementation, testing, deployment, and documentation were completed by the development team.

Project Cost

The total development cost was calculated based on the monthly salary of the team members involved in the project. With 4 team members working for the estimated project duration, the overall manpower cost formed the major part of the project expenditure.

<u>Additional Expense</u>

- Barcode Reader Cost = ₹1,600

Additional hardware costs included the purchase of a barcode reader used for scanning student ID cards during attendance recording.

<u>Total Estimated Cost:</u> ₹4,13,800

# CHAPTER 5: IMPLEMENTATION & APPLICATION

## 5.1 Implementation

### 5.1.1 Development Environment Setup

| Component | Details |
|---|---|
| Operating System | Windows 11 |
| Web Server | Apache 2.4.58 (via XAMPP v8.2.4) |
| Server-Side Language | PHP 8.2.4 |
| Database | MySQL 8.0.32 |
| Front-End Languages | HTML5, CSS3, JavaScript |
| IDE | Visual Studio Code |

### 5.1.2 Database Implementation

The MySQL database named "lcms" was created using phpMyAdmin. The core table creation SQL:

**For Administrator:**

```
CREATE TABLE `administrator` (
  `aid` int(11) NOT NULL,
  `afname` varchar(255) DEFAULT NULL,
  `alname` varchar(255) DEFAULT NULL,
  `email_id` varchar(255) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
```

**For Attendance:**

```
CREATE TABLE `attendance` (
  `att_id` int(11) NOT NULL,
  `rollno` varchar(45) DEFAULT NULL,
  `status` varchar(10) DEFAULT NULL,
  `date` date DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
```

**For Department:**

```
CREATE TABLE `dept` (
  `did` int(11) NOT NULL,
  `dname` varchar(255) DEFAULT NULL,
  `status` enum('0','1') NOT NULL DEFAULT '1'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
```

**For Entry Log:**

```
CREATE TABLE `entry_log` (
  `entry_log_id` int(11) NOT NULL,
  `rollno` varchar(45) NOT NULL,
  `log_time` time NOT NULL,
  `is_late` tinyint(1) DEFAULT NULL,
  `log_date` date DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
```

**For Gatekeeper:**

```
CREATE TABLE `gatekeeper` (
  `gid` int(11) NOT NULL,
  `gfname` varchar(255) DEFAULT NULL,
  `glname` varchar(255) DEFAULT NULL,
  `email_id` varchar(255) DEFAULT NULL,
  `hire_date` date DEFAULT NULL,
  `status` enum('0','1') DEFAULT '1',
  `gaddress` varchar(255) DEFAULT NULL,
  `gmobile` varchar(255) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
```

**For No ID:**

```
CREATE TABLE `no_id` (
  `no_id` int(11) NOT NULL,
  `rollno` varchar(20) NOT NULL,
  `log_date` date NOT NULL,
  `log_time` time NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
```

**For Section In-charge:**

```
CREATE TABLE `section_incharge` (
  `si_id` int(11) NOT NULL,
  `did` int(11) DEFAULT NULL,
  `year` year(4) DEFAULT year(curdate()),
  `tid` int(11) DEFAULT NULL,
  `academic_year` enum('1st','2nd','3rd') DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
```

**For Student:**

```
CREATE TABLE `student` (
  `sid` int(11) NOT NULL,
  `fname` varchar(255) DEFAULT NULL,
  `mname` varchar(255) DEFAULT NULL,
  `lname` varchar(255) DEFAULT NULL,
  `rollno` varchar(255) NOT NULL,
  `did` int(11) DEFAULT NULL,
  `fathername` varchar(255) DEFAULT NULL,
  `mothername` varchar(255) DEFAULT NULL,
  `mobile` varchar(255) DEFAULT NULL,
  `email_id` varchar(255) NOT NULL,
  `pwd` varchar(255) NOT NULL,
```

```
  `gender` varchar(255) DEFAULT NULL,
  `semester` varchar(50) DEFAULT NULL,
  `sem_year` varchar(4) DEFAULT NULL,
  `log_date` varchar(25) DEFAULT 'curdate()',
  `log_time` varchar(25) DEFAULT 'curtime()'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

**For Teacher:**

```
CREATE TABLE `teacher` (
  `tid` int(11) NOT NULL,
  `tfname` varchar(255) DEFAULT NULL,
  `tlname` varchar(255) DEFAULT NULL,
  `email_id` varchar(255) DEFAULT NULL,
  `did` int(11) DEFAULT NULL,
  `tmobile` varchar(15) DEFAULT NULL,
  `hire_date` date DEFAULT NULL,
  `taddress` varchar(255) DEFAULT NULL,
  `status` enum('0','1') NOT NULL DEFAULT '1',
  `hod` enum('Yes','No') NOT NULL DEFAULT 'No',
  `tmname` varchar(255) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

**For User:**

```
CREATE TABLE `user` (
  `email_id` varchar(255) DEFAULT NULL,
  `pwd` varchar(255) DEFAULT NULL,
  `role` varchar(255) DEFAULT NULL,
  `log_date` date DEFAULT NULL,
  `uid` int(11) NOT NULL,
  `log_time` varchar(50) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

### 5.1.3 User Authentication Module

The User Authentication Module is one of the most important security components of the Late Comers Management System (LCMS). This module is responsible for verifying user identity and providing secure access to the system based on user roles and permissions. It ensures that only authorized users can log in and access the functionalities assigned to them, thereby protecting sensitive attendance records and administrative data from unauthorized access.

The authentication system was implemented using PHP with secure password handling techniques. User passwords are stored in encrypted form within the database using bcrypt password hashing, which provides strong protection against password theft and unauthorized access. Instead of storing passwords as plain text, the system securely hashes passwords before saving them in the database. This significantly improves overall system security and reduces the risk of data compromise.

During the login process, the user enters their email ID or username along with their password through the Login Page. The system then retrieves the encrypted password stored in the database and uses PHP's built-in password_verify() function to compare the entered password with the hashed password securely. Authentication is successful only when the entered password matches the encrypted password stored in the database. If incorrect credentials are entered, the system denies access and displays an appropriate error message.

After successful authentication, the system automatically creates a secure user session. Important user information such as user ID, username, email ID, and role is stored in PHP session variables. These session variables help maintain the logged-in state of the user throughout the application and prevent repeated authentication requests while navigating different modules of the system.

The User Authentication Module also implements role-based access control to ensure that users can only access features and functionalities permitted for their specific role. Roles such as Admin, Faculty, Student, and Gatekeeper are assigned during user registration or account creation. Based on the assigned role, users are redirected to their respective dashboards after successful login.

To improve security further, role verification checks are placed at the beginning of every protected PHP page within the system. These checks verify whether the currently logged-in user has permission to access the requested module or functionality. If unauthorized access is attempted, the system restricts access and redirects the user appropriately. This mechanism helps prevent unauthorized users from accessing confidential attendance records, administrative controls, or restricted system operations.

The module also supports session management features such as login validation, session maintenance, and secure logout functionality. When a user logs out, the session variables are destroyed to ensure that no unauthorized access can occur from the same browser session.

Overall, the User Authentication Module plays a vital role in maintaining the security, privacy, and reliability of the Late Comers Management System. By implementing secure password encryption, session management, and role-based access control, the module ensures safe and controlled system usage for all categories of users within the institution.

### 5.1.4 Barcode Scanning Module

The Barcode Scanning Module is one of the core functional components of the Late Comers Management System (LCMS). This module was developed to provide fast, accurate, and automated attendance recording at the institution entry point. It is specifically designed for use by gatekeepers to simplify the attendance management process and eliminate the delays and errors associated with manual attendance entry.

The scanning interface contains a dedicated text input field with the autofocus attribute enabled. This feature ensures that the cursor remains automatically active inside the input field at all times, allowing continuous barcode scanning without requiring the gatekeeper to click the input field repeatedly. This design significantly improves operational efficiency during busy entry periods when a large number of students arrive simultaneously.

The barcode scanner used in the system functions similarly to a keyboard input device. When a student ID card barcode is scanned, the scanner automatically enters the barcode data into the active input field as text. Immediately after entering the barcode value, the scanner automatically triggers an Enter key press event. This mechanism enables quick and seamless transfer of scanned information into the system without any manual typing.

JavaScript event listeners are implemented within the scanning interface to detect the Enter key action generated after barcode scanning. Once the Enter key event is detected, the barcode value is captured instantly and processed automatically. The scanned data is then sent to the PHP backend server using AJAX (Asynchronous JavaScript and XML) technology.

The use of AJAX technology allows attendance data to be transmitted and processed in real time without reloading or refreshing the webpage. This improves system responsiveness, reduces waiting time, and provides a smoother user experience for gatekeepers. After processing the scanned barcode, the system immediately retrieves student information from the database and records the attendance entry automatically.

The Barcode Scanning Module also supports automatic latecomer detection by comparing the student entry time with predefined institutional attendance timings. Based on the result, the system identifies whether the student arrived on time or late and updates the attendance status accordingly. The confirmation panel displayed after scanning provides instant feedback by showing details such as student name, roll number, entry time, and attendance condition.

This module significantly improves the efficiency, speed, and reliability of attendance recording within the institution. It minimizes human errors, eliminates manual paperwork, and supports real-time attendance monitoring. By integrating barcode scanning with AJAX-based real-time processing, the system ensures accurate attendance management while enhancing overall user convenience and operational performance.

**Key Code Snippet Late Detection Function (PHP):**

The following PHP function was implemented in the Late Comers Management System (LCMS) to automatically determine whether a student is late based on the institution's predefined attendance threshold time. This function compares the student's entry time with the specified cutoff time and returns a value indicating whether the student arrived late or on time.

```
function isLate($entry_time) {
The
   $threshold = "08:45:00";
   return (strtotime($entry_time) >strtotime($threshold)) ?1 : 0;
}
```

In this function, the variable $threshold stores the predefined attendance cutoff time, which is set to **08:45:00 AM**. This time acts as the standard reporting limit defined by the institution. Any student entering after this time is considered a latecomer by the system.

The PHP built-in function strtotime() is used to convert both the student's entry time and the threshold time into Unix timestamp format. Converting the time values into timestamps makes it easier and more accurate to compare time values programmatically.

The function then uses a conditional (ternary) operator to check whether the student's entry time is greater than the threshold time. If the condition is true, the function returns the value 1, which indicates that the student is late. If the condition is false, the function returns 0, indicating that the student arrived on time.

This late detection function plays a vital role in automating attendance processing within the LCMS. It eliminates the need for manual verification of late arrivals and ensures accurate, fast, and consistent attendance evaluation. The function is integrated with the barcode scanning and attendance recording modules, enabling real-time identification of latecomers immediately after barcode scanning.

By automating late attendance detection, the system improves efficiency, reduces human error, and supports effective monitoring of student punctuality within the institution.

**Key Code Snippet Barcode Processing (PHP AJAX Handler):**

The following PHP code snippet was implemented in the Late Comers Management System (LCMS) to process barcode data received from the barcode scanning interface through AJAX requests. This module handles student identification, attendance verification, late detection, and attendance record processing in real time.

```
$barcode = $_POST["rollno"];
$query = "SELECT * FROM student WHERE rollno_value= ?";
$stmt = $conn->prepare($query);
$stmt->bind_param("s", $rollno);
$stmt->execute();
$result = $stmt->get_result();
if ($result->num_rows === 0) {
   echo json_encode(["status"=>"error","msg"=>"Student not found"]);
   exit;
}
$student = $result->fetch_assoc();
$entry_time = date("H:i:s");
$is_late = isLate($entry_time);
// Insert attendance record and check notification threshold...
```

In this code, the barcode value scanned from the student ID card is received from the AJAX request using the $_POST["rollno"] variable. The scanned barcode data is then used to search for the corresponding student record in the student database table.

A prepared SQL statement is used to execute the database query securely. Prepared statements help protect the system from SQL injection attacks and improve database security. The bind_param() function binds the barcode value to the query parameter, while the execute() function runs the query against the database.

After execution, the query result is retrieved using the get_result() method. The system then checks whether any matching student record exists in the database. If no matching student is found, the system immediately returns a JSON error response containing the message "Student not found" and terminates further execution using the exit statement. This ensures that invalid or unregistered barcode scans are handled properly.

If a valid student record is found, the student information is retrieved using the fetch_assoc() function. The current system time is then captured using the date("H:i:s") function and stored in the $entry_time variable. This entry time represents the exact time when the student entered the institution premises.

The captured entry time is passed to the isLate() function, which compares the entry time with the predefined attendance threshold time. Based on this comparison, the function determines whether the student arrived late or on time and stores the result in the $is_late variable.

After successful verification and late detection, the system proceeds to insert the attendance record into the database and checks whether the student has exceeded the late attendance notification threshold. Additional actions such as updating attendance logs, displaying confirmation messages, and generating notifications can also be performed at this stage.

This barcode processing module enables real-time attendance recording and automatic latecomer detection within the LCMS. By combining AJAX technology, PHP processing, prepared SQL statements, and automated attendance verification, the system achieves fast, secure, and efficient attendance management with minimal manual intervention.

### 5.1.5 Notification Module

The Notification Module is an important feature of the Late Comers Management System (LCMS) designed to monitor repeated late attendance behaviors and automatically inform the concerned faculty members or section in-charges. This module helps improve student punctuality by ensuring that repeated late arrivals are identified quickly and brought to the attention of responsible academic authorities for further action.

The module operates automatically whenever a new attendance record is inserted into the system. After recording a student's attendance, the system checks whether the student was marked as late by verifying the value of the is_late field in the attendance or entry log records. If the student is identified as late, a PHP function is triggered to calculate the total number of late attendance occurrences associated with that student.

The system performs a database query to count all attendance records where is_late = 1 for the particular student. This count represents the total number of times the student has arrived late to the institution. The Notification Module continuously monitors this count in real time whenever new attendance entries are processed.

A notification is automatically generated whenever the total late attendance count becomes exactly divisible by 3. In other words, notifications are created when the student reaches 3, 6, 9, 12, or any further multiples of three late entries. This condition-based notification mechanism helps avoid unnecessary alerts while still ensuring that repeated late attendance patterns are monitored effectively.

Once the notification condition is satisfied, the system automatically creates a new notification record and stores it in the notifications table within the database. The notification entry includes important details such as the student ID or roll number, notification message, attendance status, notification date, and the assigned section in-charge or faculty member responsible for monitoring the student. This relationship helps ensure that the correct faculty member receives the notification regarding the student's attendance behaviors.

The Notification Module supports better academic supervision by enabling faculty members and section in-charges to identify students with recurring punctuality issues quickly. Faculty members can then take appropriate corrective measures such as counselling, warnings, discussions with parents, or attendance monitoring to improve student discipline and punctuality.

In addition, the automated notification process eliminates the need for manual tracking of repeated latecomers, reduces administrative workload, and improves the efficiency of attendance monitoring. Since notifications are generated automatically in real time, the system ensures timely reporting and faster response to attendance-related issues.
Overall, the Notification Module enhances the effectiveness of the Late Comers Management System by supporting proactive attendance management, improving communication between faculty and students, and encouraging better punctuality and discipline within the institution.

## 5.2 Results

### 5.2.1 Performance Benchmark Results

The performance evaluation of the Late Comers Management System (LCMS) was conducted to measure the efficiency, speed, reliability, and stability of the application under different operational conditions. Various performance metrics were tested to ensure that the system could handle real-time attendance recording, user authentication, report generation, and continuous system operation effectively. The benchmark results demonstrated that the system performed efficiently and met the expected functional and performance requirements.

The barcode scanning and attendance recording process achieved an average scan-to-record response time of approximately 1.4 seconds. This indicates that the system was capable of processing barcode input, verifying student details, detecting late attendance, and storing attendance records in near real time. Even during peak operational load, the maximum scan-to-record processing time remained within 2 seconds, demonstrating stable system responsiveness under high student entry traffic conditions.

The report generation module was also evaluated to determine the time required to retrieve and process attendance data from the database. The system successfully generated attendance reports containing approximately 100 records within 50 seconds. This performance confirmed that the database and reporting modules were functioning accurately and efficiently for attendance analysis and record management purposes.

The Login Module demonstrated a fast response rate with an average login authentication time of 0.7 seconds. This result indicated that user authentication, password verification, session creation, and role-based redirection processes were executed quickly and securely, providing a smooth user experience for administrators, faculty members, students, and gatekeepers

The overall reliability and availability of the system were measured through uptime monitoring over a continuous period of 30 days. During this duration, the system maintained an uptime of approximately 99.3%, indicating high operational stability and minimal downtime. This result confirmed that the system was capable of supporting continuous attendance management activities without major interruptions or failures.

The summarized performance benchmark results are shown below:

| **Performance Metric** | **Result** |
|---|---|
| Average Barcode Scan-to-Record Time | 1.4 seconds |
| Maximum Scan-to-Record Time (peak load) | 2 seconds |
| Report Generation Time (100 records) | 50 seconds |
| Login Response Time | 0.7 seconds |
| System Uptime (30 days) | 99.3% |

These benchmark results demonstrate that the Late Comers Management System is efficient, responsive, and reliable for practical institutional use. The system successfully handled real-time attendance operations, minimized delays, and maintained stable performance throughout testing and deployment conditions.

### 5.2.2 Functional Outcomes

All major functional requirements of the Late Comers Management System (LCMS) were successfully implemented, integrated, and tested during the project development process. The system operated accurately and efficiently under different testing conditions and successfully achieved the primary objectives defined during the requirement analysis phase. The successful implementation of all modules

demonstrated the effectiveness and reliability of the proposed automated attendance management solution.
The barcode scanning module performed efficiently during system testing and practical evaluation. More than 250 barcode scan tests were conducted to verify the reliability and accuracy of the attendance recording process. During these tests, the system successfully identified students through their barcode-enabled ID cards and recorded their attendance details correctly in the database. Each scan accurately captured important information such as student identity, entry time, attendance status, and latecomer condition without producing any major errors or data inconsistencies. The real-time scanning process significantly reduced manual effort and improved attendance recording speed and accuracy.

The automatic late detection mechanism also functioned successfully with a high level of accuracy. The system automatically compared each student's entry time with the predefined institutional late attendance threshold time of **08:45 AM** using the implemented PHP late detection function. Students arriving after the specified threshold were automatically marked as late in the attendance records. This automated process eliminated the need for manual verification of late arrivals and ensured consistent and accurate attendance evaluation throughout the system.

The notification module operated effectively by monitoring repeated late attendance behaviors and generating alerts automatically when students accumulated multiple late entries. The system successfully generated notifications whenever a student reached late attendance counts divisible by three, such as 3, 6, 9, and so on. These notifications were correctly linked to the respective section in-charges and displayed through the Faculty Dashboard for monitoring and supervision purposes. This feature enabled faculty members to identify students with recurring punctuality issues and take necessary corrective actions to improve attendance discipline.

Role-based authentication and dashboard functionalities were also tested successfully. Different categories of users, including Administrators, Faculty Members, Students, and Gatekeepers, were able to access only the modules and functionalities assigned to their specific roles. Login authentication, session handling, and access control mechanisms worked accurately and securely throughout the testing process.

The attendance recording, report generation, and database management modules also performed as expected. Attendance records were stored correctly, attendance reports were generated successfully, and all database operations functioned reliably without data loss or corruption. The integration between the frontend interface, backend PHP logic, AJAX communication, and MySQL database remained stable during all testing activities.

Overall, the functional outcomes confirmed that the Late Comers Management System successfully fulfilled its intended objectives of automating attendance management, improving punctuality monitoring, reducing manual work, minimizing human error, and enhancing institutional attendance management efficiency.

### 5.2.3 Comparison with Existing System

The comparison between the existing manual attendance system and the proposed Late Comers Management System (LCMS) clearly demonstrates the improvements achieved through automation and digital attendance management. The traditional manual system relied heavily on handwritten attendance records and manual verification processes, which were time-consuming, error-prone, and difficult to manage efficiently. In contrast, the proposed LCMS introduced barcode-based attendance recording, automated late detection, real-time monitoring, and secure database management, resulting in improved speed, accuracy, security, and operational efficiency.

One of the major improvements introduced by the proposed system was the automation of attendance entry through barcode scanning. In the existing system, attendance details such as student name, roll number, department, and entry time were recorded manually in physical registers, which consumed significant time and increased the possibility of human error. The proposed LCMS replaced this process with automated barcode scanning, enabling fast and accurate attendance recording in real time.

The speed of attendance recording also improved significantly. In the manual system, recording attendance for a single student required approximately 15 seconds due to manual writing and verification. However, in the proposed LCMS, the average barcode scan-to-record processing time was reduced to approximately 1.4 seconds per student. This improvement helped eliminate long queues and delays during peak entry periods.

Accuracy was another major area of improvement. The manual attendance process was prone to errors such as incomplete entries, incorrect roll numbers, duplicate records, and missing attendance details. The proposed LCMS achieved nearly 100% attendance recording accuracy through automated barcode identification, database validation, and real-time processing mechanisms.

The existing manual system did not support real-time attendance monitoring because records were maintained only in physical registers. In contrast, the proposed LCMS provided real-time access to attendance data through interactive dashboards for administrators, faculty members, students, and gatekeepers. This allowed users to monitor attendance records instantly from the system interface.

Report generation was also significantly improved in the proposed system. In the manual process, preparing weekly attendance reports required approximately 45 minutes because records had to be searched and calculated manually. The LCMS reduced this time drastically by generating attendance reports automatically within approximately 10 seconds using database queries and automated report generation modules.

The notification process also became more efficient in the proposed system. In the existing manual system, repeated latecomers were identified manually and informed through written memos or verbal communication. The proposed LCMS introduced an automated notification mechanism that generated alerts whenever a student accumulated three or more late attendance records. These notifications were displayed automatically to the concerned section in-charges through the Faculty Dashboard.

Data security was another important improvement achieved through the proposed system. The manual attendance system relied on physical registers, which were vulnerable to damage, loss, unauthorized modification, and difficult record maintenance. The proposed LCMS stored attendance data securely in an encrypted database with role-based access control and authentication mechanisms, significantly improving data protection and reliability.

The summarized comparison between the two systems is shown below:

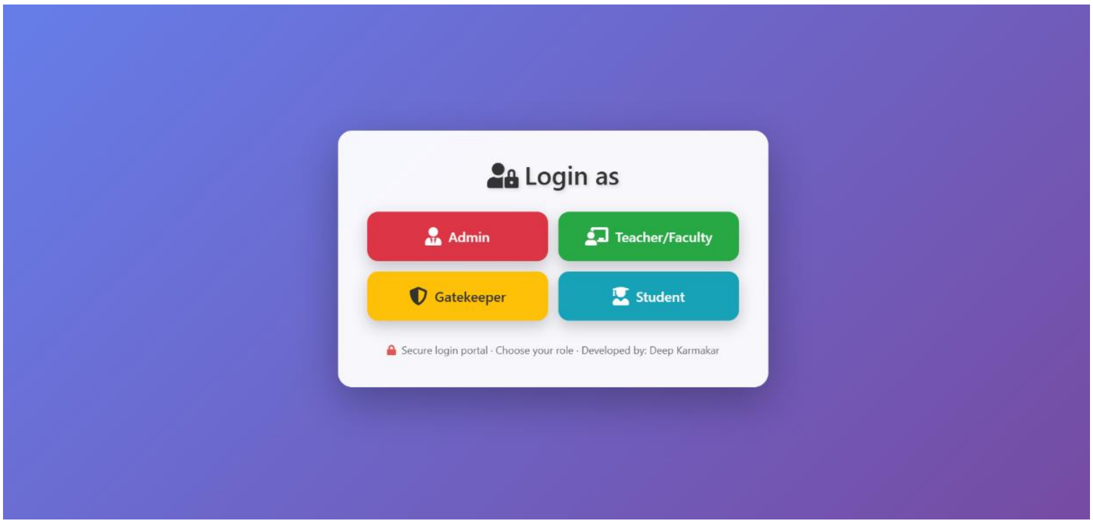
| **Parameter** | **Existing System (Manual)** | **Proposed LCMS** |
|---|---|---|
| Data Entry Method | Manual handwriting | Automated barcode scan |
| Speed | 15 seconds per student | 1.4 seconds per scan |
| Accuracy | Prone to errors | 100% accurate |
| Real-Time Access | No | Yes, via dashboard |
| Report Generation | 45 minutes/week | 10 seconds |
| Notification | Manual memo | Automatic at 3+ late arrivals |
| Data Security | Physical register | Encrypted database |

Overall, the comparison clearly shows that the proposed Late Comers Management System provided substantial improvements over the existing manual attendance management process by enhancing efficiency, accuracy, speed, security, and real-time monitoring capabilities.

### 5.2.4 Application Interfaces (Screenshots)

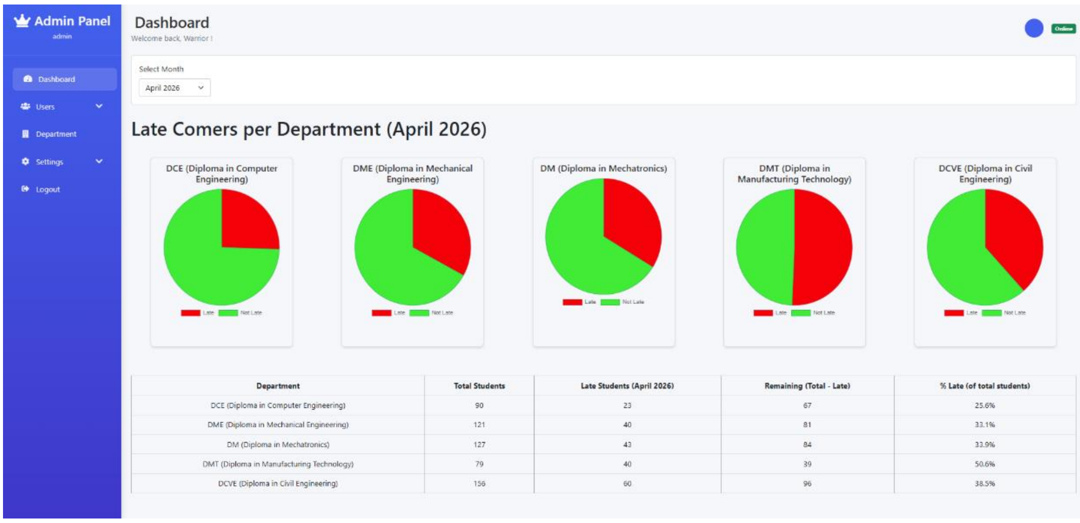
The Late Comers Management System (LCMS) was developed with multiple user-friendly interfaces to support different categories of users, including Administrators, Faculty Members, Students, and Gatekeepers. Each interface was designed according to the specific responsibilities and operational requirements of the respective users. The screenshots below represent the major application interfaces implemented within the system.

## Fig. 5.1 - Login Page:



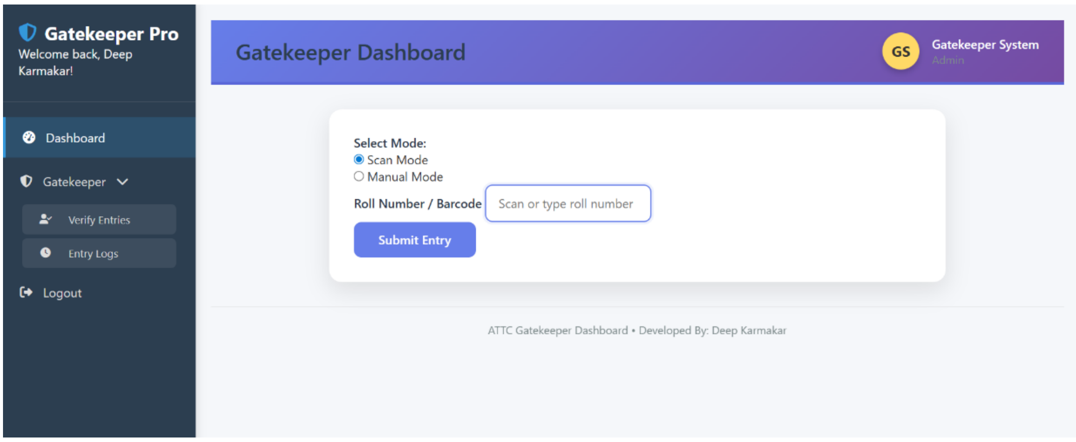

The Login Page serves as the primary authentication interface of the system. It allows authorized users to securely access the application using their login credentials. The page contains input fields for username or email ID and password, along with login and password recovery options. After successful authentication, users are redirected automatically to their respective dashboards based on their assigned roles.

## Fig. 5.2 - Admin Dashboard:



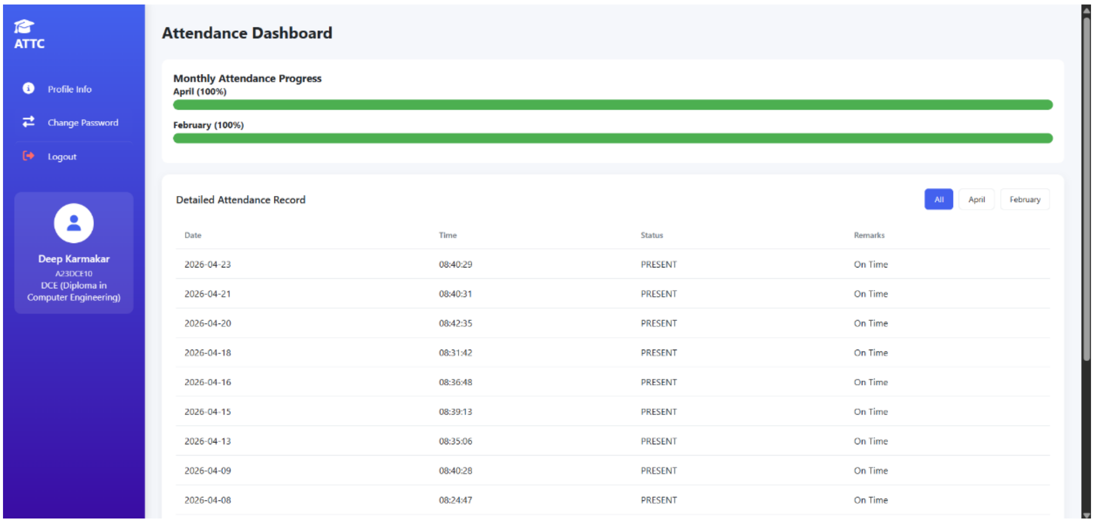

The Admin Dashboard functions as the central management interface of the LCMS. It provides administrators with complete control over system operations such as user management, student record management, section in-charge assignments, attendance monitoring, and report generation. The dashboard also displays graphical attendance statistics, summary cards, and navigation modules for efficient institutional management.

**Fig. 5.3 - Gatekeeper Dashboard:**




The Gatekeeper Dashboard is specifically designed for attendance recording through barcode scanning. It contains a barcode input field with automatic focus functionality, enabling continuous attendance scanning without manual interaction. After each scan, the system displays student details, attendance status, and late entry information instantly. The dashboard also maintains a daily entry log for monitoring student movement in real time.

**Fig. 5.4 - Student Dashboard:**




The Student Dashboard provides students with access to their attendance information and punctuality records. Students can view total late arrivals, attendance percentages, and detailed attendance history through a user-friendly interface. The dashboard also allows profile management and helps students monitor their attendance performance and punctuality status effectively.

**Fig. 5.5 - Faculty Dashboard:**


The Faculty Dashboard enables faculty members and section in-charges to monitor student attendance and latecomer activities efficiently. It provides access to attendance records, student search functionality, notifications for repeated late arrivals, and attendance reports. Faculty members can use this interface to supervise attendance behaviors and take corrective action when necessary.

These application interfaces collectively improve usability, operational efficiency, attendance monitoring, and overall system interaction within the Late Comers Management System.

# CHAPTER 6: CONCLUSION & FUTURE SCOPE

## 6.1 Conclusion

The Late Comers Management System (LCMS) was developed to address the need for a faster, more accurate, secure, and efficient method of monitoring and managing student late arrivals at the Advanced Technical Training Centre (ATTC), Bardang, East Sikkim. The project was successfully designed, implemented, tested, and validated according to the defined functional and technical requirements. The final results demonstrated that the system effectively fulfilled its intended objectives and provided a practical solution for improving institutional attendance management.

By integrating barcode technology with PHP-MySQL based web application development, the LCMS successfully replaced the traditional manual attendance recording process with an automated and reliable system. The application enabled quick and real-time recording of student entries through barcode scanning, significantly reducing the time and effort required for attendance management. The automated late detection mechanism accurately identified students arriving after the predefined threshold time and updated attendance records instantly within the database.

The system also provided secure and centralized storage of attendance records using a MySQL database, improving data organization, accessibility, and reliability compared to physical attendance registers. Attendance information could be accessed in real time by authorized users through dedicated dashboards, enabling efficient monitoring and faster decision-making. In addition, attendance reports and latecomer records could be generated automatically with minimal manual effort, improving administrative efficiency and reducing paperwork.

The role-based dashboard mechanism implemented within the LCMS ensured that different categories of users, including Administrators, Faculty Members, Students, and Gatekeepers, could access functionalities relevant to their assigned responsibilities. This improved both system security and operational management. The automated notification feature further enhanced attendance supervision by informing Section In-charges whenever students accumulated repeated late attendance records, helping faculty members take corrective measures to improve student punctuality and discipline.

From a technical perspective, the project demonstrated the effective use of open-source technologies such as XAMPP, PHP, MySQL, HTML, CSS, JavaScript, AJAX, and Bootstrap for developing a cost-effective, scalable, and user-friendly institutional management system. The successful integration of barcode scanning, AJAX-based real-time processing, secure authentication mechanisms, and database management highlighted the practical application of modern web technologies in solving real-world administrative problems.

Performance testing and functional evaluation confirmed that the system operated efficiently under different testing conditions. The barcode scanning process, automatic late detection, report generation, login authentication, and notification modules all functioned accurately and reliably throughout testing and deployment. Compared to the existing manual attendance system, the LCMS significantly improved operational speed, attendance accuracy, data security, usability, and real-time monitoring capabilities.

## 6.2 Future Scope

While the current version of the Late Comers Management System (LCMS) is fully functional and successfully achieves its defined objectives, several advanced features and improvements can be implemented in future versions to further enhance the system's efficiency, usability, scalability, and overall performance. These future enhancements would help transform the system into a more intelligent, flexible, and institution-wide attendance management solution.

One possible future enhancement is the integration of RFID (Radio Frequency Identification) or biometric authentication technologies such as fingerprint or facial recognition systems. These technologies can provide even faster, more secure, and contactless attendance recording compared to barcode scanning. Biometric authentication would also eliminate the possibility of proxy attendance and improve the overall reliability and authenticity of attendance records.

The system can also be expanded into a mobile application for Android and iOS platforms. A dedicated mobile application would allow students, faculty members, administrators, and parents to access attendance information directly through smartphones. Mobile support could provide features such as attendance notifications, report viewing, profile management, and instant communication, thereby improving accessibility and user convenience.

Another major future enhancement is cloud-based deployment of the system. Currently, the system operates on a local server environment, but migrating the application to a cloud platform would enable remote access, centralized data management, automatic backups, and improved scalability. Cloud deployment would allow multiple departments, campuses, or institutions to access and manage attendance data from a centralized platform securely and efficiently.

Advanced analytics and Artificial Intelligence (AI)-based prediction features can also be incorporated into future versions of the LCMS. Machine learning algorithms and data analytics tools could be used to identify attendance trends, analyze student punctuality behavior, predict irregular attendance patterns, and generate intelligent reports. Such features would help administrators and faculty members take proactive measures to improve student discipline and academic performance.

Additional communication features such as SMS and email notification systems can further enhance the effectiveness of attendance monitoring. Automatic alerts could be sent to parents or guardians whenever students arrive late repeatedly or remain absent. This would strengthen communication between the institution and families and encourage better student accountability and punctuality.
The LCMS can also be integrated with the institution's existing academic management or ERP (Enterprise Resource Planning) system. Integration with academic records, examination systems, timetable management, and student performance databases would create a unified institutional management platform. Real-time synchronization of attendance records with academic systems would reduce duplication of work and improve administrative efficiency.

Future versions may also include advanced reporting and visualization features such as interactive dashboards, graphical attendance analytics, monthly performance summaries, and downloadable attendance reports in Excel formats. These features would improve decision-making and help administrators monitor attendance statistics more effectively.

Security enhancements such as two-factor authentication (2FA), encrypted communication protocols, automated database backup systems, and audit logging mechanisms can also be implemented to further strengthen system security and protect sensitive attendance data from unauthorized access or data loss.

### 6.2.1 SMS Notification Integration

The current version of the Late Comers Management System (LCMS) provides in-system notifications that are displayed on the Faculty Dashboard whenever a student repeatedly arrives late to the institution. These notifications help Section In-charges monitor student punctuality and take corrective actions when necessary. However, future versions of the system can be further enhanced by integrating SMS notification services to provide faster and more direct communication.

The proposed SMS Notification Integration feature would allow the system to automatically send real-time text message alerts to Section In-charges, faculty members, and parents whenever a student’s late attendance count reaches a predefined threshold, such as 3, 6, or 9 late arrivals. This automation would ensure that important attendance-related information is communicated immediately without requiring users to log in to the system dashboard manually.

The integration can be implemented using SMS gateway services such as Twilio or Indian SMS gateway providers that support API-based message delivery. Through these services, the LCMS would be able to generate and send SMS notifications automatically whenever the notification module detects repeated late attendance behaviors. The system can dynamically include details such as student name, roll number, date, time, and total late attendance count within the message content.

This feature would significantly improve communication speed and responsiveness within the institution. Faculty members and Section In-charges would receive immediate alerts regarding attendance violations, enabling them to take timely disciplinary or counselling actions. Parents or guardians could also be informed instantly about their ward's punctuality status, improving parental involvement and accountability in student attendance management.

In addition to improving communication efficiency, SMS notification integration would enhance the overall effectiveness of the attendance monitoring system. Since SMS alerts can be received on any mobile phone without internet access, the feature would ensure broader accessibility and reliability compared to dashboard-only notifications. It would also reduce delays in identifying attendance issues and strengthen coordination between the institution, faculty members, and parents.

From a technical perspective, the SMS module can be integrated with the existing notification system through API-based communication between the LCMS backend and the SMS service provider. PHP scripts can automatically trigger SMS requests whenever the notification condition is satisfied, while delivery logs can be stored in the database for monitoring and verification purposes.

### 6.2.2 Mobile Application Development

In future versions of the Late Comers Management System (LCMS), a dedicated mobile application can be developed for Android and iOS platforms to further improve system accessibility, usability, and user convenience. The introduction of a mobile application would allow students, faculty members, administrators, and other authorized users to access attendance-related information directly through their smartphones from virtually anywhere.

The mobile application would provide a more flexible and user-friendly platform compared to the existing web-based interface. Students would be able to log in securely and view important information such as attendance records, late-entry history, attendance percentages, notifications, and personal profile details directly from the mobile application. This would help students monitor their punctuality and attendance status more conveniently in real time without requiring access to a desktop system.

Faculty members and Section In-charges would also benefit significantly from the mobile application. They could receive instant push notifications whenever students repeatedly arrive late or cross predefined attendance thresholds. Faculty users would also be able to monitor attendance records, view latecomer reports, search student information, and track attendance activities directly through their mobile devices. This would improve responsiveness and allow faculty members to supervise attendance more effectively even when away from their workstations.

The proposed mobile application would communicate with the LCMS backend server through a RESTful API (Application Programming Interface) layer developed using the existing PHP infrastructure. The REST API would act as an intermediate communication layer between the mobile application and the MySQL database, enabling secure and efficient data exchange. API endpoints would handle operations such as user authentication, attendance retrieval, notification management, attendance updates, and report access.

Using RESTful API architecture would also make the system more scalable and maintainable because both the web application and mobile application could share the same backend services and database structure. This approach would reduce duplication of development effort and simplify future system enhancements and maintenance activities.

The mobile application could additionally support advanced features such as biometric login authentication, QR code scanning, real-time attendance synchronization, offline data caching, and cloud-based notifications. Integration with Firebase or similar push notification services could enable instant message delivery for attendance alerts and important announcements.

From a usability perspective, the mobile application would provide a more interactive and responsive experience through simplified navigation, touch-based controls, and mobile-optimized interfaces. This enhancement would significantly improve user engagement and accessibility, especially for students and faculty members who rely heavily on smartphones for daily communication and academic activities.

### 6.2.3 Biometric Authentication Integration

As an alternative or supplementary technology to barcode scanning, future versions of the Late Comers Management System (LCMS) can incorporate biometric authentication methods such as fingerprint scanning or facial recognition for student identification and attendance recording at the institution gate. The integration of biometric technology would significantly improve the security, reliability, and authenticity of the attendance management process.

In the current system, attendance is recorded using barcode-enabled student identification cards. Although barcode scanning provides fast and automated attendance recording, certain limitations still exist, such as damaged barcode cards, lost ID cards, or the possibility of students exchanging ID cards with others. Biometric authentication would eliminate these issues by verifying unique physical characteristics of each student before attendance is recorded.

Fingerprint recognition systems can identify students based on their unique fingerprint patterns, while facial recognition systems can authenticate students automatically using camera-based image analysis. Since biometric features are unique to every individual, attendance can only be recorded for the actual student, thereby preventing proxy attendance, unauthorized entry, and false attendance records. This would significantly improve the accuracy and integrity of the attendance management system.

The biometric verification process can be integrated directly with the existing LCMS backend infrastructure. During attendance recording, the biometric device would capture the student’s fingerprint or facial image and compare it with previously stored biometric templates in the database. Once authentication is successful, the system would automatically record the student’s attendance, detect late arrival status, and update attendance records in real time.

Biometric integration would also improve operational efficiency at the institution gate. Modern biometric devices are capable of authenticating users within seconds, enabling faster attendance processing during peak entry periods. Facial recognition systems, in particular, can support contactless attendance recording, which improves convenience and hygiene while reducing physical interaction with devices.

From a security perspective, biometric authentication provides a higher level of protection compared to barcode or card-based systems because biometric traits cannot be easily duplicated, shared, or forged. This enhancement would strengthen institutional attendance monitoring and improve trust in attendance records maintained by the system.

The integration of biometric technology can further be expanded with advanced features such as real-time facial recognition monitoring, anti-spoofing mechanisms, cloud-based biometric storage, and AI-powered identity verification. Such technologies would make the LCMS more intelligent, scalable, and suitable for large educational institutions or organizations requiring highly secure attendance management systems**.**

### 6.2.4 QR Code Upgrade

Future versions of the Late Comers Management System (LCMS) can enhance the existing barcode-based attendance mechanism by upgrading it to a QR (Quick Response) code-based identification system. QR codes provide several advantages over traditional linear barcodes, including higher data storage capacity, faster scanning capability, and improved flexibility in attendance management operations.

Unlike standard barcodes, which can store only limited information in a horizontal format, QR codes are two-dimensional and capable of storing significantly larger amounts of data. This would allow additional student-related information such as student ID, roll number, department, semester, and encrypted authentication details to be securely embedded within a single QR code. The increased data capacity would improve identification accuracy and support more advanced attendance verification processes.

Another major advantage of QR code integration is the ability to perform scanning through mobile device cameras. In the current barcode-based system, attendance recording depends primarily on dedicated USB barcode scanners connected to the Gatekeeper's workstation. With QR code technology, smartphones, tablets, or other camera-enabled devices could be used for attendance scanning, reducing the dependency on specialized hardware devices.

This enhancement would make the attendance system more flexible, portable, and cost-effective. Gatekeepers or authorized staff members could use mobile devices to scan student QR codes directly through a mobile application or web-based scanning interface. Such portability would be especially useful during outdoor activities, events, temporary entry points, or situations where dedicated scanning hardware is unavailable.

QR code technology also provides faster and more reliable scanning performance compared to traditional barcodes. Since QR codes can be scanned from different angles and even when partially damaged, the attendance recording process would become more efficient and resilient under practical operating conditions. This would help reduce scanning delays and improve student entry management during peak attendance hours.

From a security perspective, QR codes can also support encrypted or dynamically generated data, making unauthorized duplication or misuse more difficult. Additional authentication measures such as time-sensitive QR codes, digital signatures, or encrypted tokens could further strengthen attendance security and prevent fake attendance attempts.

The QR code-based system could be integrated seamlessly with the existing LCMS backend infrastructure. The attendance recording workflow, database management, late detection mechanism, and notification modules would continue to operate similarly while replacing the barcode scanning input with QR code scanning functionality.

In addition, QR code integration would improve the scalability of the system. Since mobile devices can act as scanning tools, the institution could expand attendance monitoring to multiple entry points or departments without requiring additional barcode scanning hardware investments. This would make the system more adaptable for larger institutions and future institutional expansion.

### 6.2.5 Cloud Deployment

Future versions of the Late Comers Management System (LCMS) can be migrated from the current local XAMPP server environment to a cloud-hosted infrastructure such as Amazon Web Services, Google Cloud, or other shared hosting and cloud service platforms. Cloud deployment would significantly improve the accessibility, scalability, reliability, and long-term maintainability of the attendance management system.

In the current implementation, the LCMS operates within a local server environment, meaning that system access is generally limited to devices connected to the institution's internal network. By deploying the system on a cloud platform, authorized users such as administrators, faculty members, students, and gatekeepers would be able to access the system securely from any location through the internet using computers, tablets, or smartphones.

Cloud deployment would enable administrators and faculty members to monitor attendance records, generate attendance reports, review latecomer statistics, and manage attendance activities remotely without depending on the institution's local network infrastructure. This would provide greater operational flexibility and improve accessibility, especially for institutions operating across multiple departments, campuses, or remote locations.

Another major advantage of cloud hosting is improved scalability. As the number of users, attendance records, and institutional operations increases over time, cloud infrastructure can easily accommodate higher workloads by allocating additional computing resources dynamically. This would ensure smooth system performance even during periods of heavy attendance activity or increased database usage.

Cloud-based deployment would also improve system reliability and uptime. Professional cloud service providers offer high-availability infrastructure, server redundancy, and automated load balancing mechanisms that help minimize downtime and maintain continuous system operation. This would make the LCMS more dependable for daily attendance management activities.

In addition, cloud hosting provides advanced backup and disaster recovery capabilities. Automated database backups, secure cloud storage, and recovery mechanisms would help protect attendance records and user data from accidental deletion, hardware failure, or system crashes. This significantly reduces the risk of data loss and improves overall system resilience and business continuity.

Security can also be enhanced in a cloud-hosted environment through features such as encrypted communication protocols (HTTPS/SSL), firewall protection, role-based access management, server monitoring, and automated security updates. These measures would strengthen data protection and improve the overall security of the attendance management system.

The migration to cloud infrastructure can be implemented gradually using the existing PHP and MySQL architecture of the LCMS. The backend application can be deployed on cloud web servers while the database can be hosted using managed cloud database services. Additional technologies such as cloud APIs, containerization, and virtual servers can further improve deployment efficiency and scalability.

### 6.2.6 Analytics Dashboard Enhancement

Future versions of the Late Comers Management System (LCMS) can incorporate an advanced analytics and reporting module to provide deeper insights into student attendance behaviors, punctuality trends, and overall attendance performance. This enhancement would transform the system from a basic attendance management platform into a more intelligent and data-driven decision-support system for institutional administration.

The proposed analytics module could include interactive dashboards with graphical representations such as bar charts, pie charts, line graphs, heat maps, and attendance trend visualizations. These visual tools would help administrators, faculty members, and Section In-charges understand attendance behaviors more clearly and identify important patterns quickly without manually analyzing raw attendance records.

Using the analytics dashboard, users could monitor daily, weekly, monthly, or semester-wise attendance statistics and compare punctuality trends across different departments, semesters, or student groups. The system could also generate reports highlighting the most frequent latecomers, attendance percentages, peak late-entry timings, and department-wise attendance performance. Such analytical insights would improve attendance monitoring and institutional decision-making processes.

Future versions may also integrate predictive modelling and Artificial Intelligence (AI)-based analysis features. By analyzing historical attendance records and behavioral patterns, the system could identify students who may be at risk of academic performance issues, disciplinary concerns, or chronic absenteeism due to repeated late arrivals. Machine learning algorithms could evaluate attendance trends and generate early warning indicators for students requiring additional supervision or intervention.

These predictive features would allow faculty members and administrators to take proactive measures such as counselling sessions, mentoring programs, parent communication, or disciplinary guidance before attendance problems become severe. Early identification of attendance-related issues could contribute positively to student discipline, academic performance, and institutional management.

The analytics module could additionally support customizable filters and report generation options. Administrators may generate attendance reports based on date ranges, departments, semesters, student categories, or punctuality status. Downloadable reports in formats such as PDF, Excel, or CSV could also be supported for academic review and administrative documentation purposes.

From a technical perspective, the analytics dashboard can be implemented using modern frontend charting libraries and JavaScript visualization frameworks integrated with the existing PHP and MySQL backend infrastructure. Attendance data stored in the database can be processed dynamically to generate real-time visual reports and analytical summaries.

This enhancement would significantly improve institutional decision-making capabilities by converting attendance data into meaningful insights and actionable information. Faculty members would gain better visibility into student behaviors, while administrators could monitor institutional attendance performance more effectively through centralized analytical tools.

### 6.2.7 Parent Portal

A dedicated Parent Portal can be introduced in future versions of the Late Comers Management System (LCMS) to strengthen communication and collaboration between the institution and student families. This feature would allow parents or guardians to monitor their ward's attendance records, punctuality status, and late-entry history directly through a secure online interface.

The Parent Portal would provide authorized guardians with access to important attendance-related information such as daily attendance records, total late arrivals, attendance percentages, notification history, and attendance trend summaries. By providing real-time visibility into student attendance behaviors, the system would improve transparency and help parents remain actively involved in monitoring their child's academic discipline and punctuality.

One of the key functionalities of the Parent Portal would be the automated notification system. Whenever a student repeatedly arrives late or crosses predefined late-entry thresholds, the system could automatically send alerts or notifications to the registered parent or guardian. These notifications could be delivered through SMS, email, mobile push notifications, or directly within the parent portal dashboard. This real-time communication mechanism would ensure that parents are informed immediately about attendance-related concerns.

The portal could also include graphical attendance summaries and detailed reports that allow parents to track long-term attendance behaviors more effectively. Features such as monthly attendance reports, punctuality statistics, and attendance comparison charts would help guardians understand patterns in their child's attendance performance and identify areas requiring improvement.

By enabling parents to monitor attendance activities directly, the system would encourage greater accountability among students. Students are more likely to maintain punctuality and discipline when both faculty members and parents are actively aware of their attendance behaviors. This collaborative monitoring approach would help improve student responsibility and support better academic discipline within the institution.

The Parent Portal can also support additional communication features such as messaging systems, counselling notifications, meeting schedules, and academic announcements. Faculty members or administrators could communicate directly with parents regarding attendance concerns, disciplinary actions, or student progress through the portal interface.

From a technical perspective, the Parent Portal could be integrated into the existing LCMS architecture using role-based authentication and secure access control mechanisms. Parents would receive unique login credentials linked specifically to their ward's attendance records, ensuring privacy and secure access to student information. The portal could be made available both as a web-based interface and through future mobile application integration for greater accessibility.

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# APPENDICES

## Appendix A: Installation and Deployment Guide

### Step 1: Install XAMPP

Download and install XAMPP from https://www.apachefriends.org/. Launch the XAMPP Control Panel and start the Apache and MySQL services.XAMPP provides the required Apache web server, PHP runtime environment, and MySQL database services needed for running the LCMS application.

After installation, open the XAMPP Control Panel and start the following services:

- Apache
- MySQL

Ensure that both services are running successfully before proceeding to the next step. Successful startup of these services indicates that the local server environment is ready for deployment.

### Step 2: Deploy the Application Files

Copy the LCMS project folder (e.g., "lcms") to the XAMPP htdocs directory (typically C:\xampp\htdocs\ on Windows).Locate the LCMS project folder containing all application files and directories. Copy the complete project folder (for example, lcms) into the XAMPP htdocs directory.

The default path for the htdocs folder on Windows systems is:
C:\xampp\htdocs\

After copying the project folder, the final application path should appear similar to:
C:\xampp\htdocs\lcms
This step makes the application accessible through the local Apache web server.

### Step 3: Import the Database

Open phpMyAdmin at http://localhost/phpmyadmin. Create a database named "lcms". Import the provided SQL dump file (lcms.sql).

Inside phpMyAdmin:

1. Create a new database named:
   lcms
2. Select the newly created lcms database.
3. Click on the **Import** tab.
4. Choose the provided SQL database dump file:
   lcms.sql
5. Click the **Go** button to import all required database tables and records into the MySQL database system.

Successful import of the SQL file will automatically create all necessary tables such as student, teacher, attendance, user, entry_log, administrator, and other related tables required for system operation.

### Step 4: Configure Database Connection

Open lcms/config/connect.php and verify that the host, username, password, and database name match your XAMPP MySQL configuration.Verify and update the database connection settings according to the local MySQL configuration used in XAMPP. The configuration details typically include:

- Host Name
- MySQL Username
- MySQL Password
- Database Name

Example configuration:
$host = "localhost";
$username = "root";
$password = "";
$database = "lcms";

Ensure that these values match the MySQL server settings configured in the XAMPP environment.

**Step 5: Access the System**

Navigate to http://localhost/lcms in your web browser. Use default admin credentials (thewarrior16842@gmail.com / 123456) to log in.The LCMS Login Page will appear, allowing authorized users to access the system.

Use the default administrator login credentials below for the initial system login:

- Email ID:
  thewarrior16842@gmail.com
- Password:
  123456

After successful authentication, the administrator dashboard and system modules can be accessed according to user roles and permissions.

**Appendix B: Sample SQL Queries**

**Query 1: Retrieve all late entries for today**

```
SELECT s.roll_no, s.name, s.section, a.entry_time
FROM attendance_log a
JOIN students s ON a.student_id = s.student_id
WHERE a.scan_date = CURDATE() AND a.is_late = 1
ORDER BY a.entry_time ASC;
```

**Query 2: Count late arrivals per student for notification check**

```
SELECT student_id, COUNT(*) AS late_count
FROM attendance_log
WHERE is_late = 1
GROUP BY student_id
HAVING late_count % 3 = 0;
```

## Appendix C: Gantt Chart

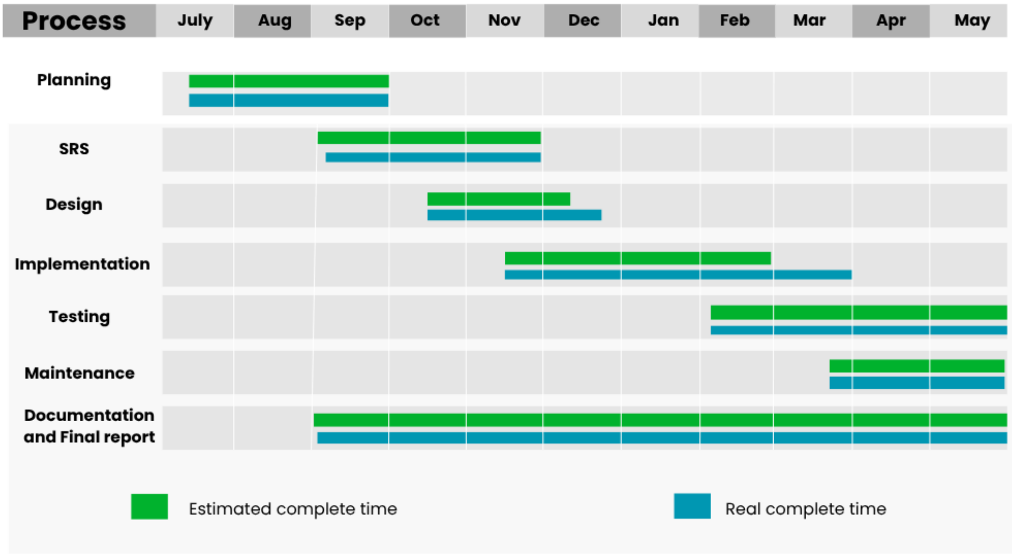


**Fig 6.1 Gantt chart**

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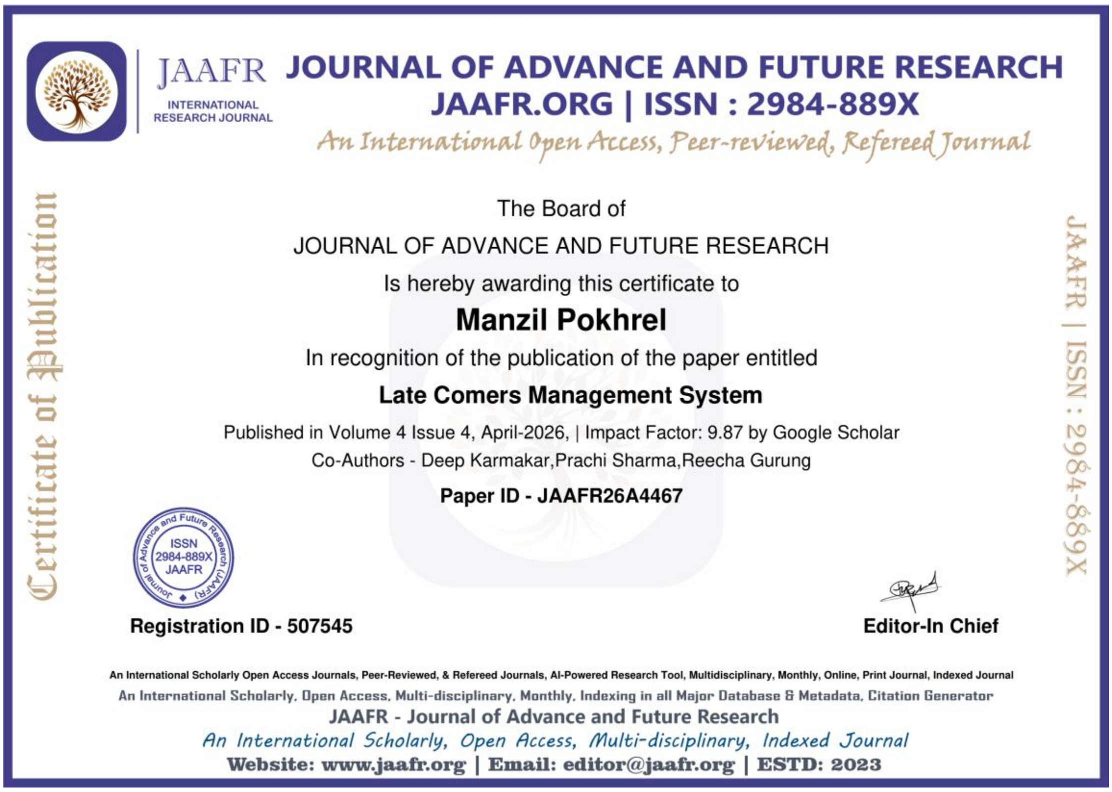

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